STRUCTURE BASED DRUG DESIGNING AND MOLECULAR DOCKING ON INFLUENZA VIRUS

ABSTRACT

Due to the mutation and drug resistance, influenza is still a major global health problem and the development of effective anti-influenza agents should be solved. Herein, we present the currents study in which a SBDD strategy was used to screen and identify novel inhibitors against crucial proteins of the influenza virus including hemagglutinin (HA), neuraminidase (NA) and M2 proton channel. The protein structures (PDB IDs: 1HGE, 1NNB and 3BKD) were retrieved and prepared for computational studies. Based on an initial set of filtering criteria, compound 5988 (sucrose) was chosen from a ligand library that had been assembled and screened. We performed molecular docking to analyze binding affinity and interaction patterns of the best ligand with its target proteins. The interactions were further assessed through docking analysis, which showed stable complex formation with hydrogen donors and acceptors present in specific amino acid residues like lysine (LYS), threonine (THR) and phenylalanine (PHE) that indicate favourable binding.

Assessments of the drug-likeness and other pharmacokinetic properties of the chosen compound were performed in SwissADME, indicating satisfactory ADMET characteristics (Table S2). Moreover, the stability and flexibility of protein–ligand complexes were determined by using molecular dynamics (MD) simulations on the iMODS server. Deformability, eigenvalues and covariance analysis supported stable structure and low flexibility of the complexes. In summary, the present study reveals sucrose to be a promising lead compound against viral proteins of influenza virus and also highlights the efficacy of integrated computational approaches toward discovering antiviral drugs. Nevertheless, additional experimental validation is necessary to establish its potential use in therapy.

Keywords: Influenza virus; Structure-Based Drug Design (SBDD); Molecular Docking; Hemagglutinin (HA); Neuraminidase (NA); M2 Protein; Molecular Dynamics Simulation; SwissADME; Protein–Ligand Interaction; Antiviral Drug Discovery

INTRODUCTION

Structure-Based Drug Design (SBDD) is a rational and efficient method in modern drug discovery, which takes advantage of the three-dimensional structures of biological targets to design or optimize therapeutic molecules. This approach utilizes structural data from techniques like X-ray crystallography and nuclear magnetic resonance spectroscopy to identify active binding sites, which can inform ligand optimization. Molecular docking (MD) is an integral part of such approaches, which predicts the probable orientation of a ligand in a protein's active site and estimates binding affinity. It can be used for virtual screening of billions compound libraries, allowing high specificity selectivity identification of potential drug candidates all in a much less time- and cost-intensive way than performed by traditional experimental approaches [1,2].

Influenza virus (IV) is a very contagious respiratory pathogen in the Orthomyxoviridae family and is associated with seasonal epidemics and global pandemics. In the past, Influenza has caused numerous pandemics — including Spanish flu in 1918, Asian flu in 1957, Hong Kong flu in 1968 and H1N1 pandemic in 2009 with millions of fatalities globally. The virus has a segmented, negative-sense single-stranded RNA genome with eight segments that encode various structural and non-structural proteins. Because of its high mutation rate, it can exhibit antigenic drift and antigenic shift, which allows the virus to escape from the host immune response and represents a major obstacle to effective therapeutic options or vaccine development [3,4].

There are three types of influenza viruses: Influenza A, B and C; however, the most virulent strains that can cause pandemics is type A. Subtypes within the family are classified by surface glycoproteins known as hemagglutinin (HA) and neuraminidase (NA); H1N1 and H3N2, for example. Lineage of the fluB genes (a group of viruses that infect only mammals) is limited to humans, chiefly responsible for annual epidemics, while type C viruses typically cause mild illness. The influenza virus is an enveloped virus made up of a lipid membrane with embedded glycoproteins (GP) and internal matrix proteins and ribonucleoprotein complexes [3].

Among the viral proteins that have been identified are those with critical roles in the viral life cycle, such as hemagglutinin (HA), neuraminidase (NA), and the M2 proton channel (M2) that spans the virus membrane. Hemagglutinin is a trimeric glycoprotein that

mediates the binding of viral agent to receptor on host cell and membrane fusion. Neuraminidase, a tetrameric enzyme that cleaves sialic acid residues from the surface of host cells preventing viral aggregation and thereby enabling the release of newly produced virions. The M2 protein acts as a proton-selective ion channel that is responsible for regulating the acidification within its viral particle and this step is crucial for uncoating of the viral genome inside the host cell. As critical components of many biological functions, these proteins have great potential as targets for the development of antiviral drugs [5,6,7].

Molecular docking is an essential technique in influenza drug discovery since it allows for the selection of potential inhibitors that bind to these target proteins and inhibit their function. It is useful to analyze binding interactions, binding energies and important amino acid residues contributing to ligand binding. Before experimental validation, this computational strategy facilitates the rapid screening and prioritization of candidate molecules, accelerating drug discovery [1].

Besides docking, Normal Mode Analysis (NMA) was performed for protein stability as well as analysis of flexibility confirmed by iMODS server. In contrast to traditional molecular dynamics simulations, iMODS offers a fast and efficient computational procedure for studying the intrinsic motions and dynamic behaviour of proteins at the level of their internal coordinates. Analyses include properties like deformability, B-factor motion, eigenvalues, variance as well as covariance matrices and elastic network models. The eigenvalue provides a measure of the stiffness of the protein structure, with lower eigenvalues signifying increased flexibility and susceptibility to deformation while higher eigenvalues signify greater rigidity in the structure. And deformability plots highlight flexible regions and covariance analysis shows correlated/anti-correlated motions between two residues. Such approach is especially valuable for verification of docking data by analysing stability and dynamic behaviour of protein–ligand complexes to improve reliability of computer-aided drug design research [8,9].

This study involved the use of various bioinformatics databases and computational tools to expedite the drug development process. Protein structures were sourced from the Protein Data Bank (PDB) and ligand data was collected from PubChem and ChEMBL databases. Protein sequence and functional information were obtained from UniProt. Molecular docking using AutoDock and AutoDock Vina, and protein preparation and

visualization using UCSF Chimera were performed. The preparation of ligand library was performed using ChemMine tools whereas Drug Mint was used for stimulation of drug--likeness properties. Analysis of protein stability and flexibility was performed by the iMODS server, whereas SwissDock was used as additional docking platform [2,8].

With drug-resistant strains of influenza emerging globally, there is a growing need for the development of new antiviral agents. This strategy, using structure-based drug design in combination with molecular docking and Normal Mode Analysis, allows for the identification/validation of potential inhibitors that target key viral proteins. Together, this work provides insight into both the binding interactions and structural stability of influenza protein–ligand complexes to inform therapeutic strategies against influenza virus infections.

OBJECTIVES

- To analyze the structural properties of influenza viral proteins appeared in RCSB Protein Data Bank.
- Molecular Docking Analysis To ligand sucrose retrieval and preparation from PubChem.
- Conduct molecular docking next on swiss dock to identify interaction between sucrose and protein targets.
- UCSF Chimera for examining protein–ligand interactions (e.g. hydrogen bond and binding site interaction).
- To determine the more advantageous binding orientation by comparing the interaction of sucrose with a series of influenza proteins.
- Evaluate Sucrose as a Lead Compound for Future Antiviral Drug Discovery Studies

Databases and Tools Used in the Study

In the current study, we used a number of bioinformatics databases and computational tools for structure-based drug design and molecular docking on proteins targeting influenza virus.

Table No.1: Database and Tools and their websites

NAME OF TOOLS AND DATABASES	WEBSITES
BV-BRC	https://www.bv-brc.org/
PDB	https://www.rcsb.org/
PubChem	https://pubchem.ncbi.nlm.nih.gov/
ChemMine	http://chemmine.ucr.edu/
DrugMint	http://crdd.osdd.net/oscadd/drugmint/.
AutoDock	https://www.cgl.ucsf.edu/chimera/.
SwissDock	http://www.swissdock.ch/
UCSF Chimera	http://autodock.scripps.edu/
iMODS	http://imods.chaconlab.org/
SwissADME	http://www.swissadme.ch/

- **BV-BRC:** The Bacterial and Viral Bioinformatics Resource Centre (BV-BRC) is the integrated bioinformatics database which offers the data and analysis tools for bacterial and viral pathogens. Genomic, proteomic and functional data are available in a central platform supported by the NIAID. BV-BRC integrates data from many resources, for example, PATRIC (bacterial) and ViPR (viral pathogen), to enable study of pathogen biology and evolution as well as pathogen-host interactions. It provides access to Peterson-related data including genome annotation, protein, sequence alignment, phylogenetic analysis, comparative genomics and more. Such is making it very valuable for drug discovery and vaccine research, as it also exposes hosted datasets related to infectious diseases, (influenza viruses). Within this project, BV-BRC can also be used to gain further genomic and protein specific information of influenza virus strains to enable target identification and verification.

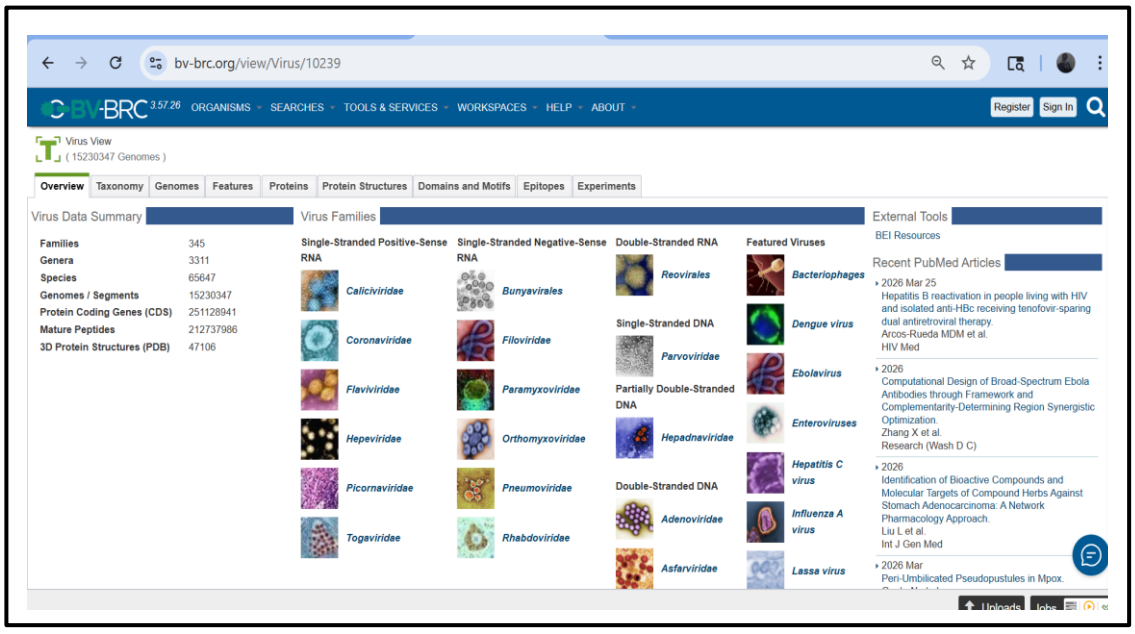


Figure No.1: BV-BRC database

- **Protein Data Bank (PDB):** The Protein Data Bank (PDB) is a worldwide open access to the three-dimensional structural data of biological macromolecules, such as proteins and nucleic acids. These are experimentally obtained using X-ray crystallography, nuclear magnetic resonance (NMR) spectrography, or cryo-electron microscopy. In this work, 3D structures of proteins such as hemagglutinin (HA), neuraminidase (NA) and

the M2 ion channel from influenza virus were fetched using this database. The structures were obtained from online database (https://www.rcsb.org/), pre-processed, and used as target proteins.



Figure No.2: Protein Databank (PDB) Database

- **PubChem:** Ligand molecules were acquired from the PubChem database for this study. The Enthusiastic public repository of chemical structures, properties, and biological activities maintained by NCBI. The ligands of interest were downloaded from: https://pubchem.ncbi.nlm.nih.gov/ in SDF/MOL formats and optimized before docking.

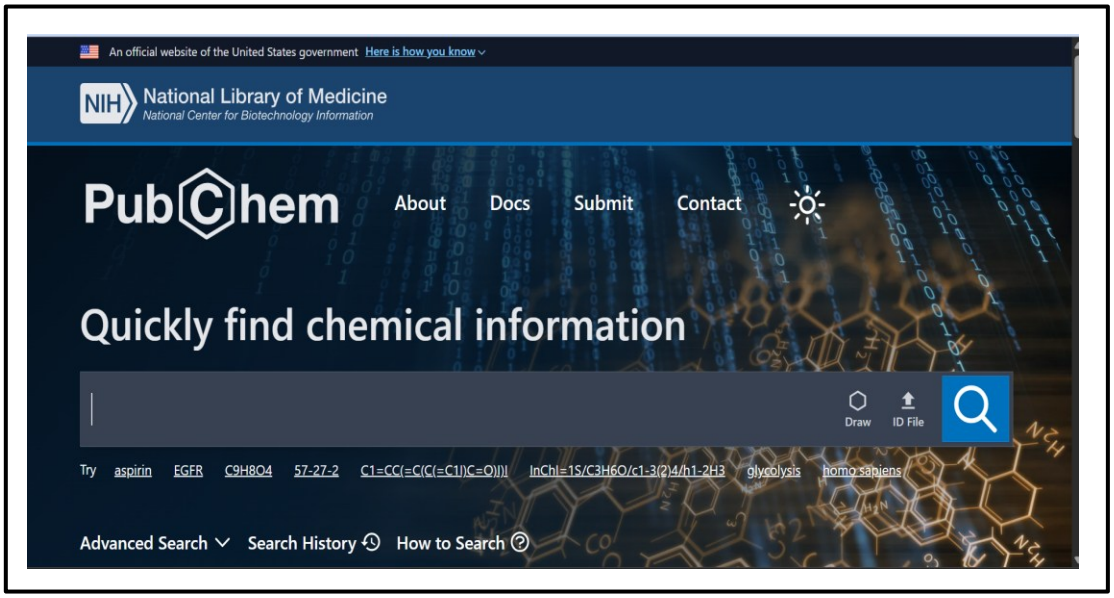


Figure NO.3: PubChem Database

- **ChemMine:** ChemMine Tools were used for ligand preparation and format conversion. It is an internet-based platform that assists with structural optimization and conversion between formats necessary for docking studies. http://chemmine.ucr.edu/ for ligand processing.

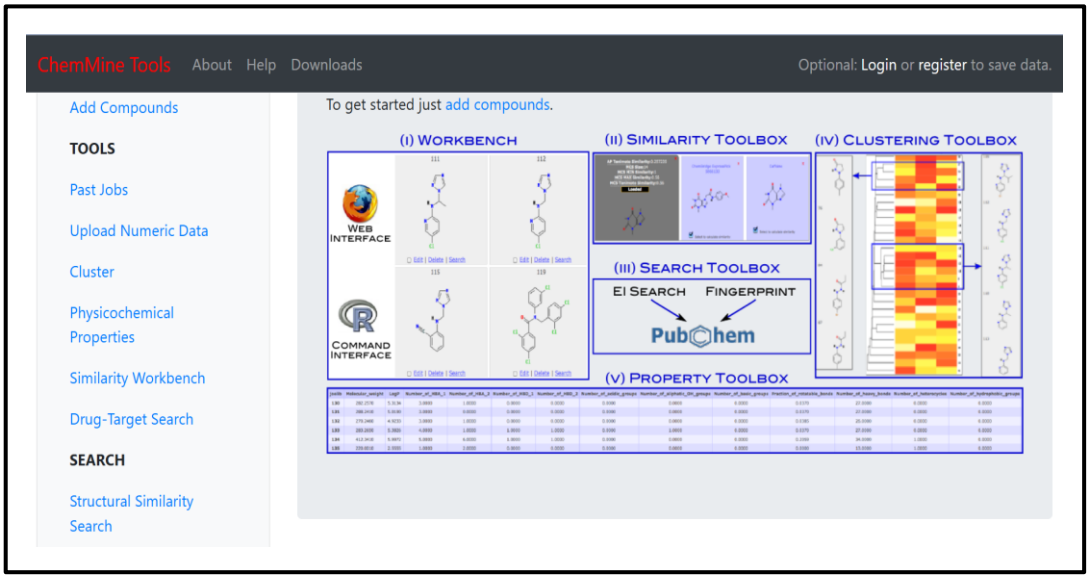


Figure No.4: ChemMine Tool

- **DrugMint:** DrugMint is freely accessible as an online web server and can be used for designing, screening, and evaluating novel compounds in computational drug discovery workflows. The tool can be accessed at: http://crdd.osdd.net/oscadd/drugmint/.

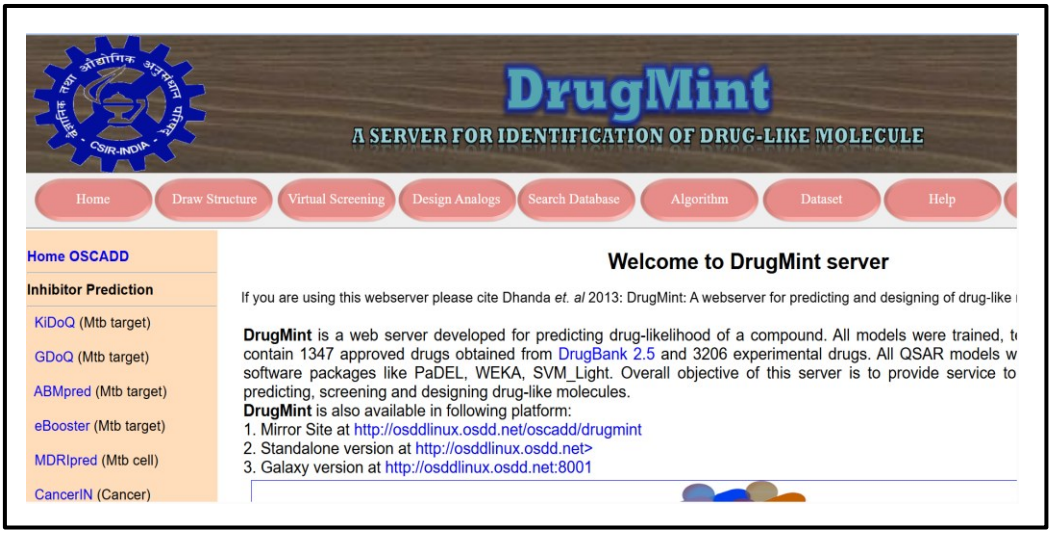


Figure NO.5: DrugMint Tool

- **AutoDock:** UCSF Chimera was used for protein preparation and visualization of the differences in molecular interactions. The software can be download at: https://www.cgl.ucsf.edu/chimera/.

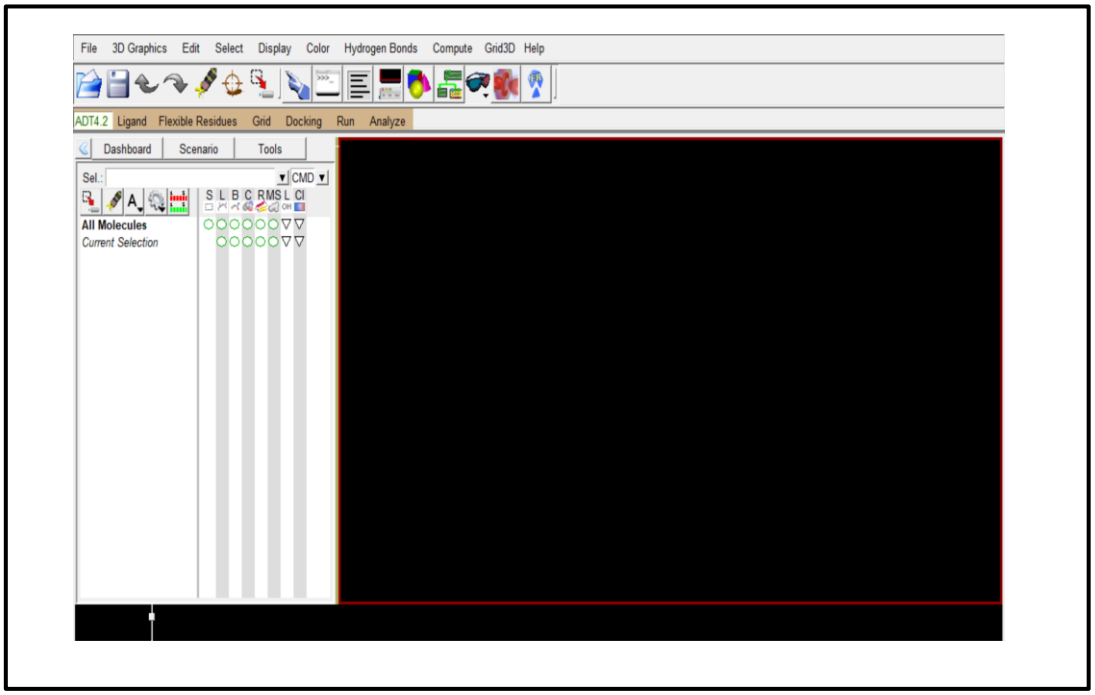


Figure No.6: AutoDock Tools

- **SwissDock:** SwissDock is a free web-based molecular docking server developed by the Swiss Institute of Bioinformatics designed for predicting small molecules and target proteins interactions. Inspired from its name, EADock DSS docking algorithm performing efficient exploration of potential binding modes of a ligand in the protein active site. SwissDock is an online docking program, which means the end-user can perform blind and site-targeted docking (without needing to be a high-performance computing expert), even without installing/setting anything on their own computer.

  The input to the tool would be a protein structure (most likely from the Protein Data Bank) and a ligand structure (available at PubChem), which are then predicted for the best binding conformations via scoring functions. It generates the output data as binding energy values (ΔG), cluster rank, and docking poses. These outcomes assist in recognizing the most stable and energetically advantageous protein–ligand connections. SwissDock also enables visualization by offering integration with the molecular

viewing tool, providing informative visualizations with detailed representations of hydrogen bonds and residue interaction sites. Molecular docking of selected ligands with the influenza virus proteins was done using SwissDock to predict their binding affinity and interaction patterns. The derived data was analyzed further to see the role of potential inhibitors. The tool can be freely accessed here: http://www.swissdock.ch/

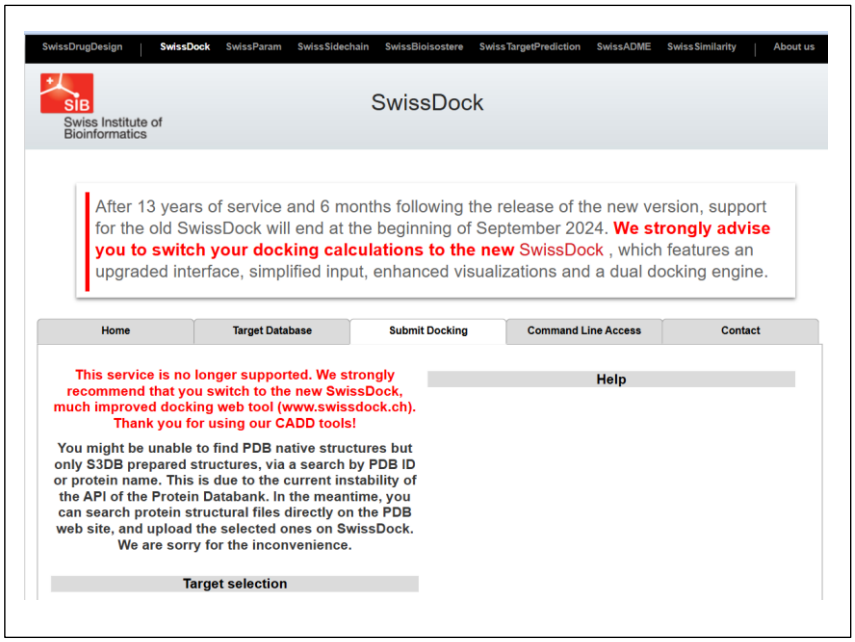


Figure No.7: SwissDock Tools

- **UCSF Chimera:** In this case, molecular docking simulations were conducted using UCSF Chimera, where the binding orientation and affinity of ligand are predicted with scoring functions. Its responsible in finding potential inhibitors. http://autodock.scripps.edu/

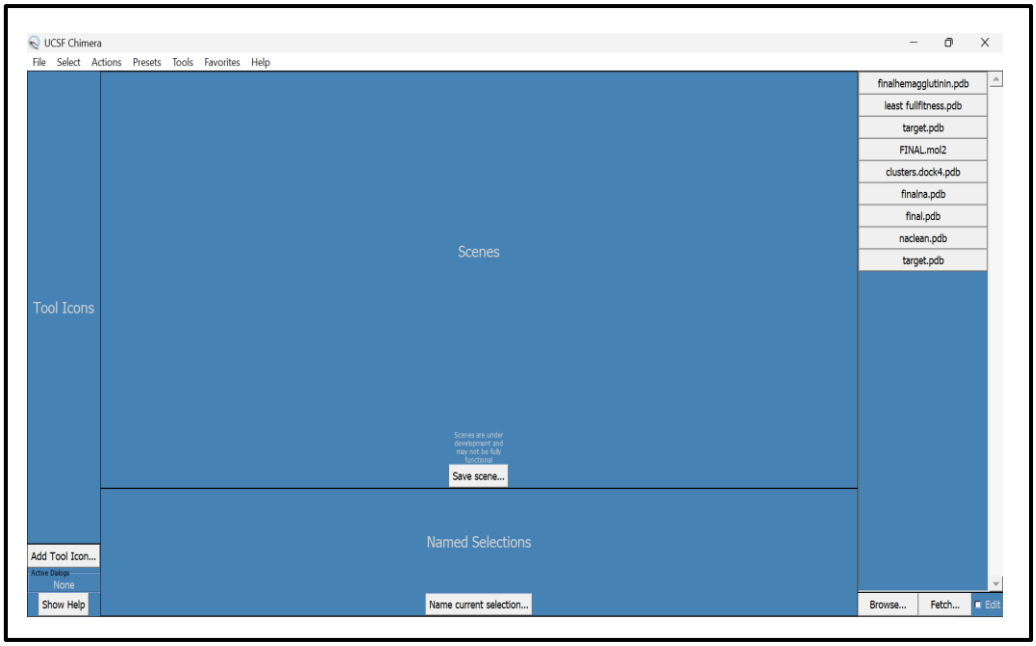


Figure No.8: UCSF Chimera Tools

- **iMODS**: Lastly, the stabilities and flexibilities of protein-ligand complexes were characterized using iMODS, which conducts Normal Mode Analysis (NMA) to study molecular motion and deformability. The tool is available at: http://imods.chaconlab.org/

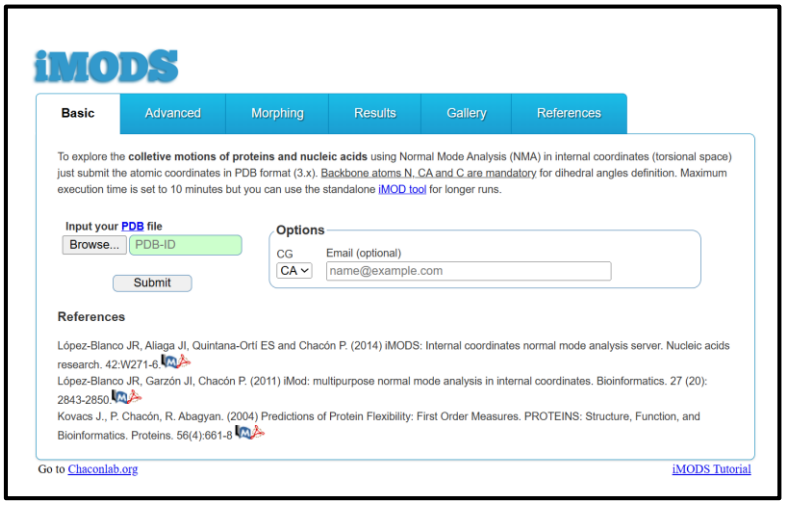


Figure No.9: iMODS Tools

- **SWISSADME:** The SwissADME is a widely used free web-based tool developed by the Swiss Institute of Bioinformatics for evaluating the pharmacokinetic properties, drug-likeness, and medicinal chemistry friendliness of small molecules. It plays a crucial role in the early stages of drug discovery by predicting key ADME parameters, including absorption, distribution, metabolism, and excretion. SwissADME provides detailed analysis such as Lipinski's Rule of Five, bioavailability score, water solubility, gastrointestinal (GI) absorption, blood-brain barrier (BBB) permeability, and P-glycoprotein substrate prediction.

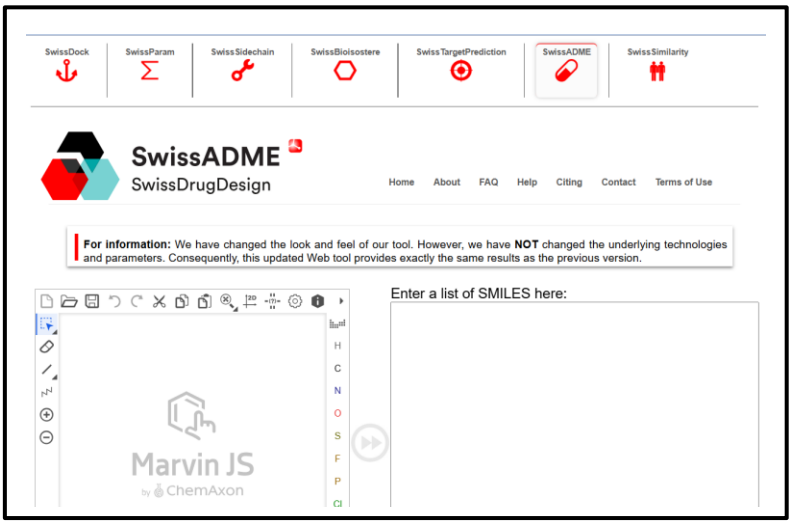


Figure No.10: SwissADME Tools

PROTEIN INFORMATION

This work involves the identification of 3 critical influenza viral proteins–Hemagglutinin (HA)(PDB ID: 1HGE), Neuraminidase (NA)–(PDB ID: 1NNB) and M2 proton channel (PDB ID: 3BKD) as molecular targets for structure-based drug design owing to their central roles during the viral life cycle along with high therapeutic relevance [10, 13].

- Hemagglutinin (HA) is a surface glycoprotein that enables the initial binding of the influenza virus to host cells. It specifically recognizes sialic acid residues on the surface of respiratory epithelial cells, enabling entry by receptor-mediated endocytosis. HA is found in a homotrimeric conformation and undergoes extensive conformational changes at low pH within the endosomes to allow for virus membrane fusion with the host cell membrane. The HA protein is depicted in its prefusion state and was the ideal structure for performing docking studies per these two inhibition assays (PDB ID: 1HGE). HA targeting also prevents viral entry, and can therefore have a protective effect at the very first stage of infection [11].
- Another important surface glycoprotein, neuraminidase (NA), promotes the release of newly produced viral particles from infected host cells. This hydrolyses terminal sialic acid residues on glycoproteins and glycolipids, which inhibits the aggregation of virions preventing propagation of infection. The structure used in this study (PDB ID: 1NNB) reveals how the active site is examined, which is highly conserved between strains of influenza. Nucleoprotein (NA) has long been an established drug target, with oseltamivir and zanamivir inhibitors created to mimic the transition state for sialic acid cleavage. NA inhibition blocks viral dissemination, which is why it represents a prime target in antiviral therapy [12,13].

- The M2 proton channel is an integral membrane protein that plays a key role during viral uncoating. Once the virus is endocytosed it works on a pH-regulated ion channel that enables protons to flow into the viral interior, which facilitates ribonucleoproteins dissociation from matrix proteins. The chosen structure (PDB ID: 3BKD) corresponds to the transmembrane domain of M2 channel tetramers. This protein has been a target of antiviral drugs including amantadine and rimantadine, but resistance mutations have made these less effective. Nonetheless, M2 is still a significant target for the creation of new inhibitors, particularly ones that are able to surpass resistance [14,15]. The three proteins together cover different and important stages in the life cycle of influenza virus, namely entry (HA), release (NA) and

uncoating (M2), so they provide a diverse multi-target basis for drug discovery. Based on their well-characterized structures, availability in the Protein Data Bank and established roles regarding viral pathogenicity, they are excellent candidates for molecular docking and simulation studies of this project [11,13].

MATERIALS AND METHODS

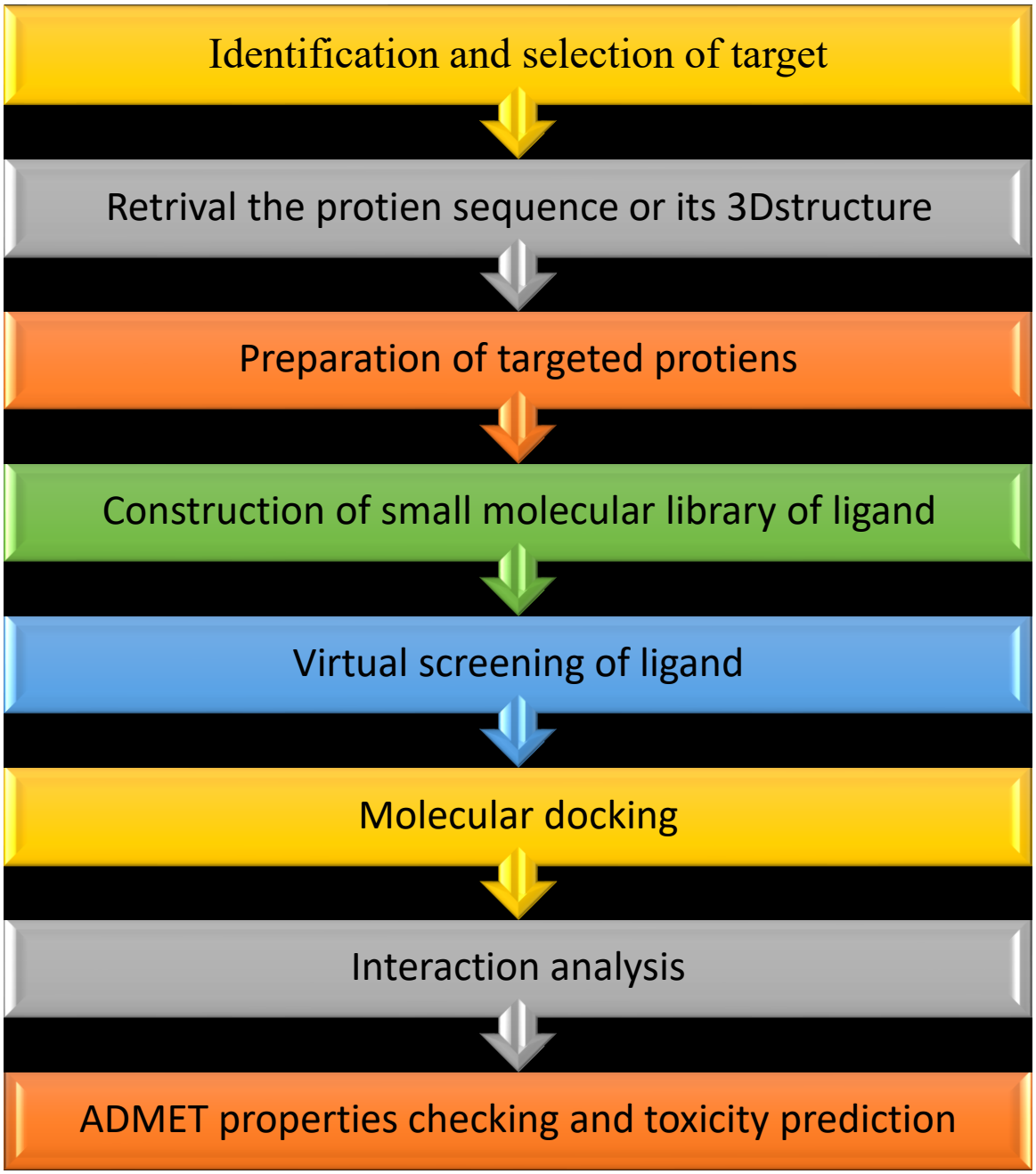


Figure No.10: pipeline of drug discovery analysis

1. Selection of Target Proteins

Selection of target proteins was based on their crucial role in the influenza virus's life cycle. The hemagglutinin (HA), neuraminidase (NA), and the M2 protein are three important proteins found in the IAV that functions within its structure: HA enables viral attachment and entry to host cells; NA releases newly formed virions, while M2 is an ion channel that has a role in uncoating. These proteins are generally regarded as important therapeutic targets for antiviral drug development purposes. Due to their essential roles, these proteins are considered prime targets for antiviral drug design [17,19].

2. Retrieval of Protein Structures and Sequences

The three-dimensional structures of target proteins under investigation were obtained from the Protein Data Bank (PDB), which is a large archive for experimentally derived macromolecular 3-d profiles. 1HGE, 1NNB, and 3BKD structures were downloaded from the PDB format. The quality of the resolution, missing residues and overall integrity of these structures were providing an insight about their use for computational studies. Accurate docking and interaction analysis require high quality structural data [16]. The three-dimensional (3D) structures of the chosen proteins were obtained from RCSB Protein Data Bank. The Protein Data Bank (PDB) IDs of the structures used in this work were:

1. Hemagglutinin (HA): 1HGE
2. M2 protein: 3BKD
3. Neuraminidase (NA): 1NNB

Additionally, for verification and analysis, protein sequences were obtained from other databases like the UniProt and NCBI. In the absence of complete structures, to predict the 3D structure homology modeling tools like SWISS-MODEL and AlphaFold can be employed.

3.Protein Preparation

Protein preparation was performed by refining the retrieved structures for molecular docking simulations. This process entailed discarding crystallographic water molecules,

co-crystallized ligands and other irrelevant heteroatoms causing hindrance in binding of the ligand. Hydrogen atoms were included in the structure of the proteins to mimic physiological conditions and appropriate charges were assigned. Subsequently, energy minimization was conducted through molecular visualization software, for example AutoDock vina, to resolve steric clashes and stabilize protein conformation. If these molecules are correctly prepared, the prediction of ligand binding interactions will also persist [23].

We conducted the following:

- Removal of water molecules (H_2O): To prevent clash with ligand binding, the water molecules were removed according to the crystallographic data.
- Addition of hydrogen atoms: Adding polar hydrogen atoms to stabilize the structure and improve docking accuracy
- Assignment of charges: These protein's Kollman charges were assigned to the prepare for docking simulations.
- Identification of active/binding sites: The identification of binding sites was made through either co-crystallized ligands or known active residues.
- Energy minimization: We energy minimized the protein structure to remove any steric clashes and obtain a low-energy conformation.

The resulting protein structures were converted to PDBQT format, which is necessary for docking with AutoDock.

4. Construction of Small Molecular Library of Ligand

A chemical ligand library was constructed using publicly available datasets. Initially, a variety of compounds were gathered and analyzed for their physicochemical and drug-likeness traits. Some of the screening criteria were Lipinski's Rule of 5, which accounts for molecular weight, lipophilicity, hydrogen bond donors and acceptors. Through systematic screening, the compound with PubChem ID 5988 was chosen as the lead ligand for further analysis based on its favourable characteristics and fit with the selected targets [21].

5. Preparation and Screening of Ligand Library

This process of repurposing screened ligands in terms of potential binding affinity toward the protein targets was carried out using virtual docking. Computational tools, including ChemMint and drug-likeness assessment platforms, were employed to filter compounds according to molecular descriptors as well as the pharmacokinetic properties. This approach aids in filtering candidates and isolating compounds with a greater likelihood of fruitful binding, thus streamlining the drug discovery process. Ligand library was designed to screen probable candidate molecules for docking studies. Chemical compounds were obtained from publicly available databases such as PubChem and further processed for virtual screening.

Library prep and screening were performed via:

- ChemMine Tools
- DrugMint

The following steps were involved:

- Small molecule data set from public chemical databases
- Proper formatting and standardizing of molecular structures
- Elimination of redundant and measurably analogous denatured compounds
- Screening based on physicochemical properties and drug-likeness criteria
- Energy minimization for ligand preparation for docking
- The compound chosen as the primary ligand from the screened library was ID 5988 due to its docking suitability and predicted drug-like properties.
- The ligand was then used for subsequent molecular docking and interaction analysis with the influenza target proteins of interest.

6. Virtual Screening and Molecular Docking

Ligand-target interaction and binding affinities prediction were executed in silico using virtual screening along with molecular docking approach. The binding interaction of the selected ligand (sucrose) with the target proteins (1HGE, 1NNB and 3BKD) were predicted by performing molecular docking studies. The docking was performed using either AutoDock or SwissDock, with scoring functions that predict interaction energies and generate the possible

binding conformations. Both preparation of the macromolecule with explicit definition of active site and setting grid parameters, to running simulations to yield multiple poses. Minimum binding energy and interaction profile were used to select the best conformer [20,22].

Docking simulations were performed with:

- SwissDock
- UCSF Chimera

The docking procedure involved:

- Generating grid box cantered on the active site
- Defining docking parameters (size and centre coordinates of the grid)
- Do docking simulations and produce several bindings conformations
- The binding energy (kcal/mol) clustering of docked poses
- The top docking pose was picked according to:
- Lowest binding energy
- Correct placement within the binding site
- Stability of interactions

7. Interaction Analysis

After docking, detailed interaction analysis was performed for the evaluation of binding mode between ligand and target proteins. For visualization of hydrogen bonds, van der Waals forces and hydrophobic interactions between the docked complexes UCSF Chimera was employed. These interactions offer insights into the stability and specificity of the ligand-protein complex, which are important for understanding how inhibition occurs molecularly.

The following interactions were studied:

- Hydrogen bond formation
- Hydrophobic interactions
- Electrostatic interactions
- Amino acid residues responsible for binding

This work used visualization techniques to help elucidate the binding mechanism, stability, and orientation of the describers complex.

8. Molecular Dynamic Simulation

Structural stability and flexibility analysis of selected influenza viral proteins i.e. Hemagglutinin (1HGE), Neuraminidase(1NNB) and M2 ion channel protein was done through molecular dynamics (MD) simulation. The iMODS server, which uses normal mode analysis (NMA) for assessing protein dynamics, performed the simulation study due to its free availability. The prepared protein structures in PDB format were then uploaded to the iMODS server and simulations were carried out with default parameters. Using NMA to predict collective motions of the proteins as a function of their internal coordinates. This process produced and examined several parameters per protein. Peaks in the deformability plots showed flexible regions in protein structures. These corresponds to the deformability plots shown in the result section of all three proteins.

Fluctuation at the residue level, which reflects atomic mobility within the protein, was investigated through B-factor plots. Then these values were compared between the proteins side-by-side to assess their relative flexibility, which can be seen from the resultant B-factor graphs. The simulated eigenvalue was applied to calculate the global stiffness of each protein structure. Lower eigenvalues denote more flexibility and higher values denote more variability. These values were used to directly infer protein stability in the results section. Variance plots were also generated to characterize how different modes contribute to protein motion. Finally, the patterns of correlated and uncorrelated as well as anti-correlated movements between pairs of amino acid residues were deduced from covariance matrices (see Results): these correspondently coloured maps allow to identify 'musical notes' played by single and pairwise residues.

Finally, elastic network models were fitted to describe interatomic interactions in respect of protein secondary and tertiary folding. The extent of the violets displayed is reflecting greater structural solidity, while a sign of greater connectivity in these models as seen within their vioplots on respective figures within results.

Generated graphs and parameters for each of these proteins were analyzed in depth and included as is, directly into the results section if they help evaluate and compare dynamic behaviour, flexibility or stability of selected proteins.

9. ADMET and Toxicity Prediction

To investigate the pharmacokinetics and toxicity characteristics of the chosen ligand, computational approaches were employed utilizing resources like SwissADME and DrugMint as the last step. The characterization of ADMET analysis should encompass absorption, distribution, metabolism, excretion, and toxicity parameters. The drug-likeness as well as safety profile of sucrose will be established in terms of

properties including the gastrointestinal absorption, blood brain barrier permeability and risk for toxicity. This step confirms the selected compound is inspiring effective yet safe for subsequent drug development studies [18].

Table No.7.1: ADMET properties of drug

Sr. No	ADMET Properties
1	Aqueous solubility
2	gastrointestinal absorption
3	Lipikinase rule
4	Pharmacokinetics

This service allows you to see if the ligand can be used in drug development studies.

RESULTS

1. Identification and Selection of Target

Target proteins were identified and selected by literature review and database analysis. Functional and genomic data for Influenza virus proteins were downloaded from the BV-BRC, a source that contains detailed information about viral proteins, including their amino acid sequence, protein structure, and biological function. According on above mentioned molecular functioning, hemagglutinin (HA), neuraminidase (NA) and matrix protein M2 were chosen as relevant drug targets due to their essential role in the cellular entry of virus, viral replication and release mechanism.

Table No.3: Target proteins selected based on functional analysis from BV-BRC.

Protein Name	Function	Importance in Influenza Virus
Hemagglutinin (HA)	Viral attachment protein	Facilitates binding and entry of virus into host cells
Neuraminidase (NA)	Viral release enzyme	Helps in release of newly formed viral particles
M2 Protein	Ion channel protein	Involved in viral uncoating and replication process

- **Hemagglutinin (HA)**

Hemagglutinin is a surface glycoprotein responsible for the attachment of the influenza virus to host cell receptors. It facilitates viral entry by binding to sialic acid residues on the host cell membrane. Due to its involvement in the initial stage of infection, HA is considered a primary target for antiviral drug development.

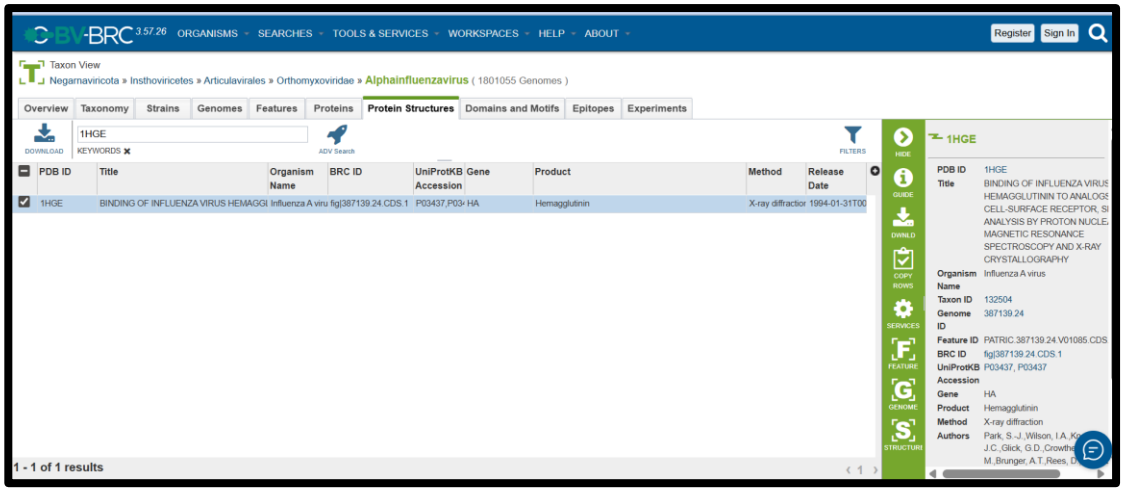


Figure No.11: selection of HA protein

- **Neuraminidase (NA)**

Neuraminidase is an enzyme that promotes the release of newly formed viral particles from infected host cells by cleaving sialic acid residues. This function is essential for viral propagation, making NA a well-established target for antiviral drugs.

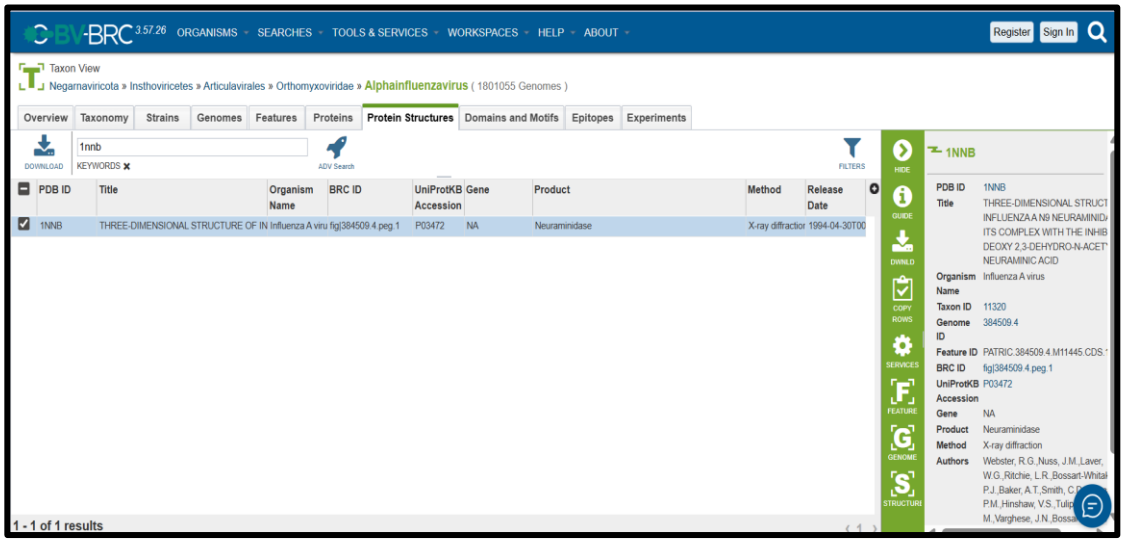


Figure No.12: selection of NA protein

- **M2 Protein**

  The M2 protein is a proton-selective ion channel that regulates the internal pH of the virus during host cell entry. It plays a critical role in viral uncoating and replication. Targeting the M2 protein can inhibit viral replication, making it an important therapeutic target**.**

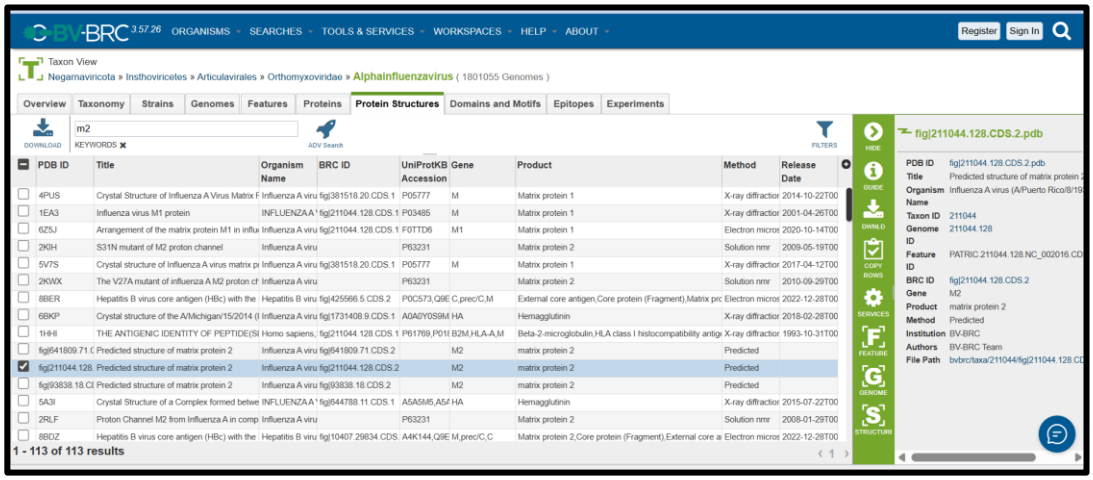


Figure No.13: selection of M2 Protein matrix

2. Retrieval of Protein Sequence and 3D Structure

The 3D structures of the identified proteins were downloaded from RCSB Protein Data Bank. These proteins have the following PDB IDs: 1HGE (Hemagglutinin), 1NNB (Neuraminidase), and 3BKD (M2 protein). these structures were chosen as they have high resolution and are structurally complete for docking analysis.

Table No.4: 3D structures of selected Influenza virus proteins retrieved from RCSB PDB

PDB ID	Protein	Resolution	Method
1HGE	Hemagglutinin	2.60 Å	X-ray crystallography
1NNB	Neuraminidase	2.80 Å	X-ray crystallography
3BKD	M2 Protein	2.05 Å	NMR / X-ray

- **Hemagglutinin (1HGE)**

  The 3D structure of Hemagglutinin (HA) reveals a trimeric glycoprotein present on the viral surface. It plays a crucial role in viral attachment by binding to sialic acid receptors on host cells, facilitating viral entry. Due to its key role in the initial stage of infection, HA is considered an important target for antiviral drug design.



Figure No.14: 3D structure and sequence of HA protein

- **Neuraminidase (1NNB)**

  The Neuraminidase (NA) structure is also a tetrameric enzyme that cleaves sialic acid residues off the host cell surface. This process allows newly infected virions to bud off from infected cells. Viral spreading can be blocked by inhibiting NA, which has made it a well-known target for drugs.

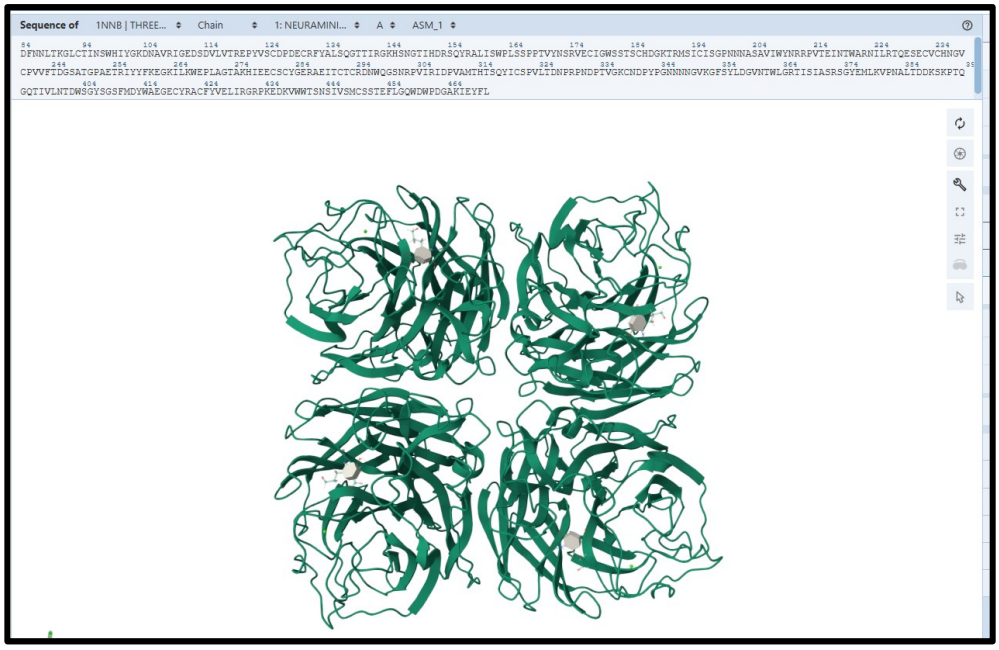


Figure No.15: 3D structure and sequence of NA protein

- **M2 Protein (3BKD)**

  The M2 protein structure is a proton-selective ion channel that resides in the viral membrane. After being directly injected into the host cell, it is critical for viral uncoating by modulating the virus's internal ph. M2 is also a target in antiviral therapy due to its functional importance in viral replication.

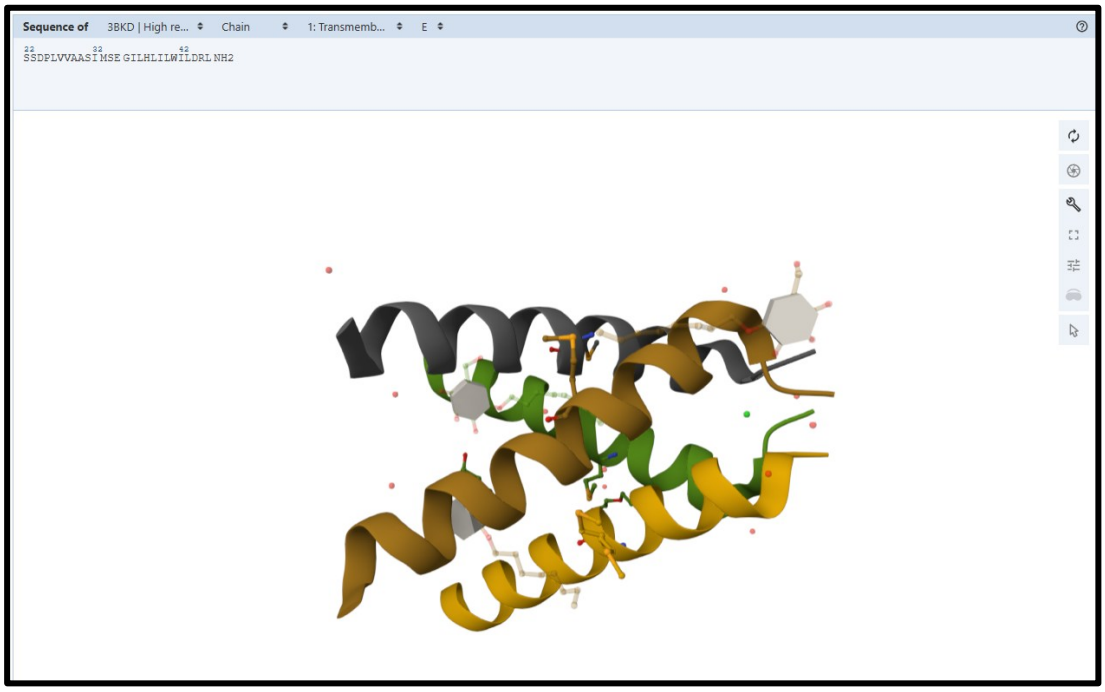


Figure No.16: 3D structure and sequence of M2 protein matrix

3. Preparation of targeted protein

Three target proteins for the study, Hemagglutinin (PDB ID: 1HGE), Neuraminidase (PDB ID: 1NNB), and M2 ion channel protein (PDB ID: 3BKD) were downloaded in PDB format from RCSB Protein Data Bank.

Preparation of protein structure for molecular docking Each protein structure was carefully prepared to ensure structural accuracy and stability before docking. All co-crystallized ligands, water molecules, and other heteroatoms that could interfere with docking analysis were removed during the preparation process. Hydrogen atoms missing from the protein structures generated by AlphaFold were added to ensure proper valency and that hydrogen bonding interactions are optimized. Energy minimization was carried out to alleviate steric clashes and stabilize the protein conformation. Furthermore, Kollman charges were applied and proteins transformed into the dockable format (PDBQT) with suitable tools.

The resulting protein structures were visually checked with molecular visualization software to validate that they had remained structurally viable and folded properly. The remaining proteins were validated to be appropriate for docking studies.

Table No.5: Protein Preparation

Protein	PDB ID	Preparation Steps Applied	Final Format
Hemagglutinin	**1HGE**	**Water removal, H-addition, energy minimization**	**PDBQT**
Neuraminidase	**1NNB**	**Water removal, H-addition, energy minimization**	**PDBQT**
M2 protein	**3BKD**	**Water removal, H-addition, energy minimization**	**PDBQT**

The original structure of Hemagglutinin (1HGE) obtained from the RCSB Protein Data Bank contained water molecules and heteroatoms. After preparation, these components were removed, and hydrogen atoms were added, resulting in a clean and optimized structure suitable for docking.

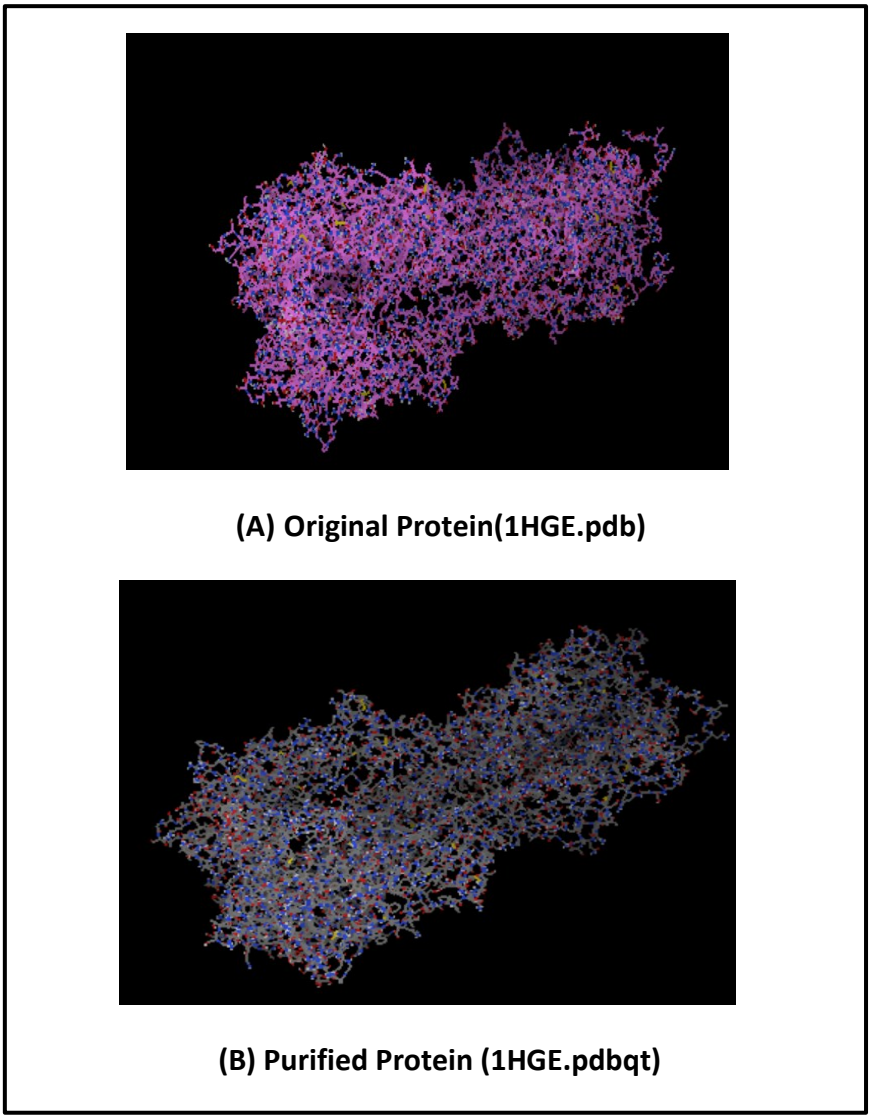


Figure No.17: Protein Preparation of Hemagglutinin (1HGE)

The original structure of Neuraminidase (1NNB) obtained from the RCSB Protein Data Bank contained water molecules and heteroatoms. After preparation, these components were removed, and hydrogen atoms were added, resulting in a clean and optimized structure suitable for docking.

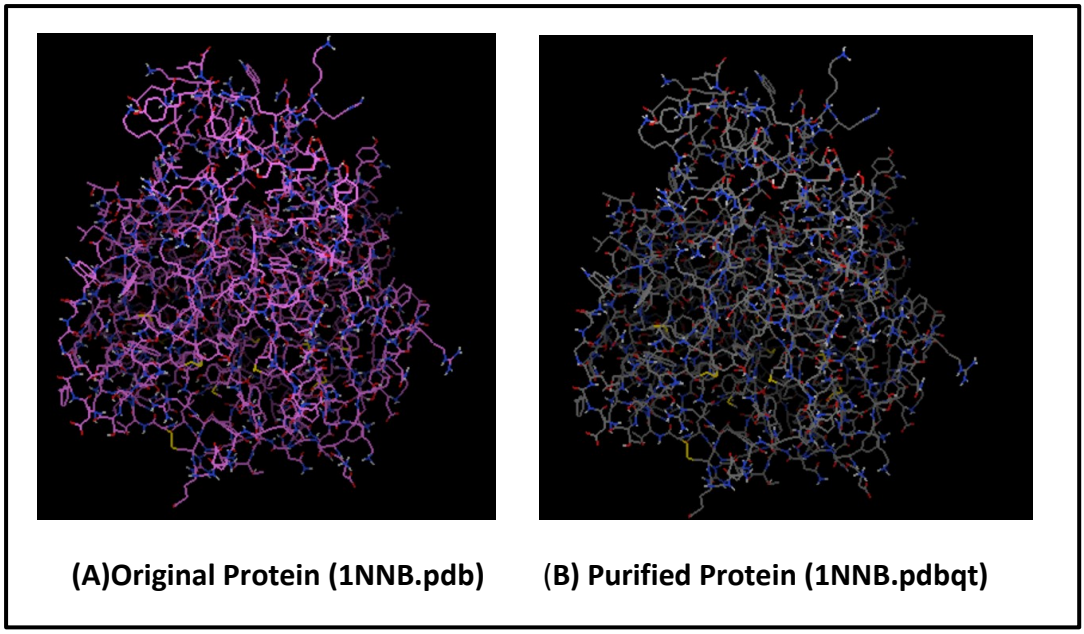


Figure No.17: Protein Preparation of Neuraminidase (1NNB)

The original structure of M2 protein (3BKD) obtained from the RCSB Protein Data Bank contained water molecules and heteroatoms. After preparation, these components were removed, and hydrogen atoms were added, resulting in a clean and optimized structure suitable for docking.

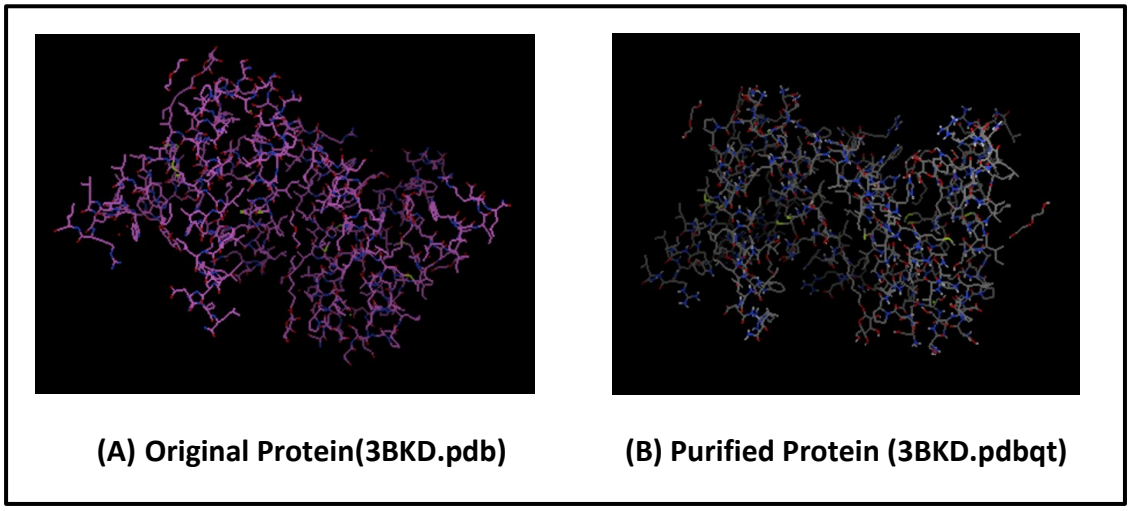


Figure No.17: Protein Preparation of M2 protein (3BKD)

4. Construction of small molecular library of ligand

For potential ligand candidates for molecular docking in future studies, ChemMine tools platform was used to generate a small molecular library. The ligands chosen for library construction were N-acetylglucosamine (NAG) and B-octyl glucoside (BOG), both recognized to have interactions with viral proteins and used generally in structural and functional analysis. A structural similarity search was conducted on ChemMine to create individual libraries of NAG (N-acetylglucosamine, 1) and BOG (n-butyl octyl glucoside, 2) ligands. These libraries consisted of compounds that shared chemical scaffolds and physicochemical properties with the parent molecules. The second step involved comparing the library for common compounds between the two datasets via an overlap analysis.

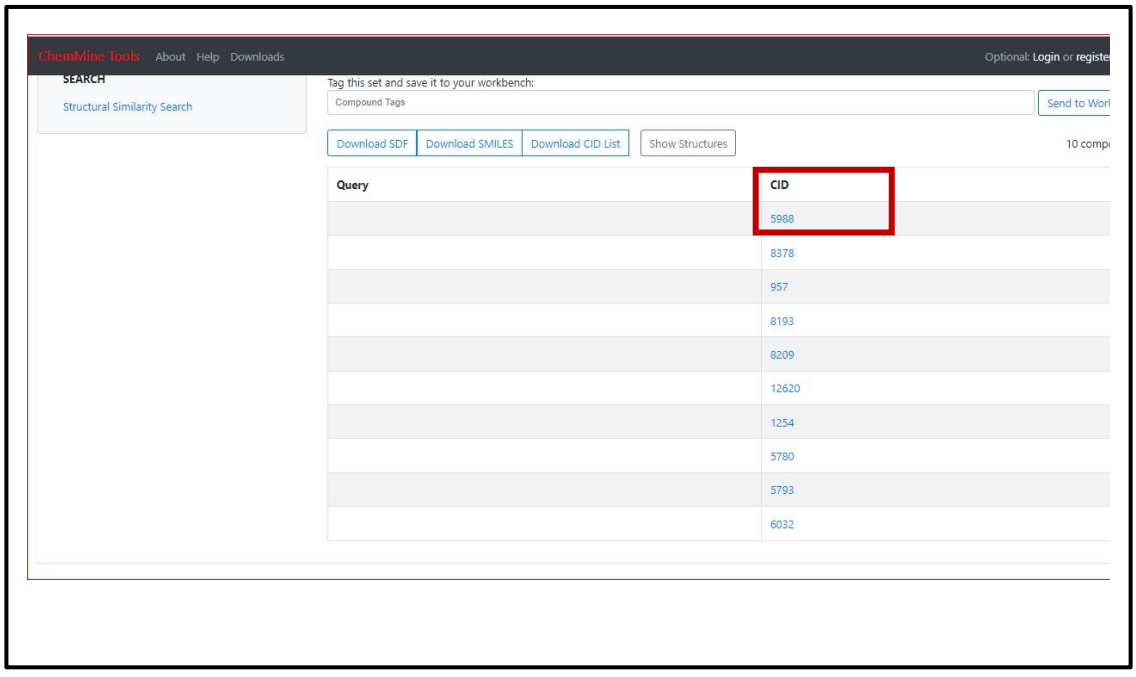


Figure No.18: Library preparation of BOG (n-butyl octyl glucoside) ligand using ChemMine

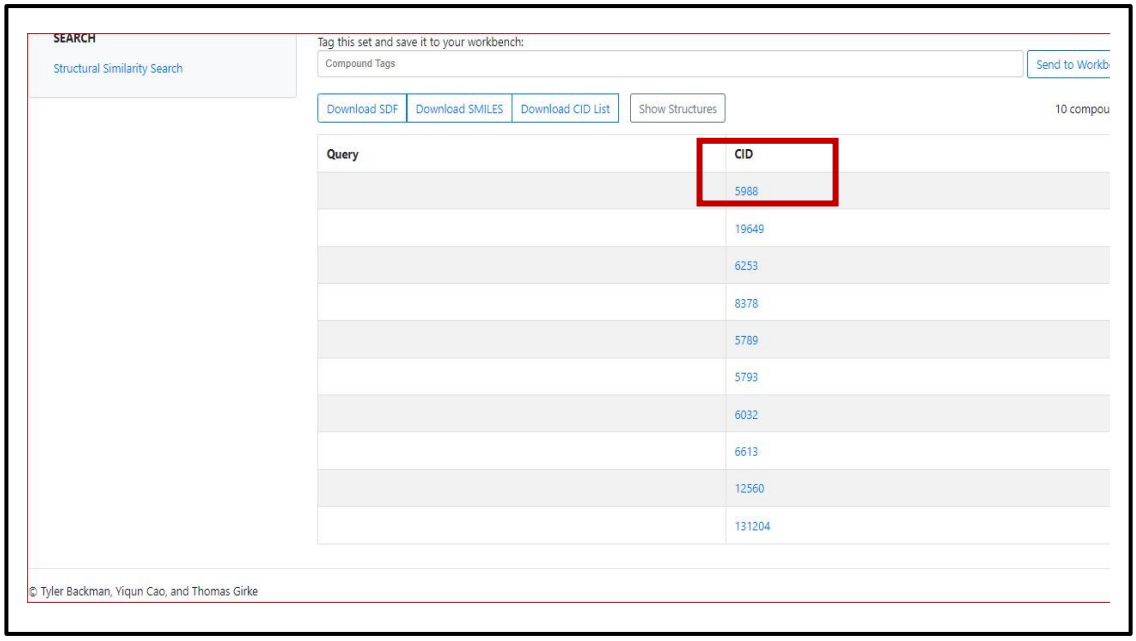


Figure No.19: Library preparation of NAG (N-acetylglucosamine) ligand using ChemMine

The comparative analysis yielded 5988 common compounds as potentially structurally conserved molecules with similar binding ability towards the selected Influenza target proteins. They provided a high-quality ligand data set for additional virtual screening and molecular docking studies.

Table No.6: Summary of generated libraries

Ligand Source	Method Used	Outcome
NAG	Structural similarity search	Library generated
BOG	Structural similarity search	Library generated
Overlap Analysis	Comparative screening	5988 common compounds identified

Ligand library construction and overlap identification workflow is shown in (Figure 18,19). In (Table 6) also is described a summary of the generated libraries and common compound numbers.

5. Virtual Screening of Ligands for Drug-Likeness Using DrugMint

The library of 5988 compounds generated based on analysis of structural similarity and overlap between NAG and BOG ligands was screened for drug-likeness using the DrugMint server. This tool uses a SVM model which is based on machine learning–based prediction, and can classify compounds as drug-like or non-drug-like according to the physicochemical and structural properties.

The ability of any of the compounds to meet drug-likeness was assessed. An extensive screening process led to the selection of top-ranking compounds based on a high probability score regarding drug-value.

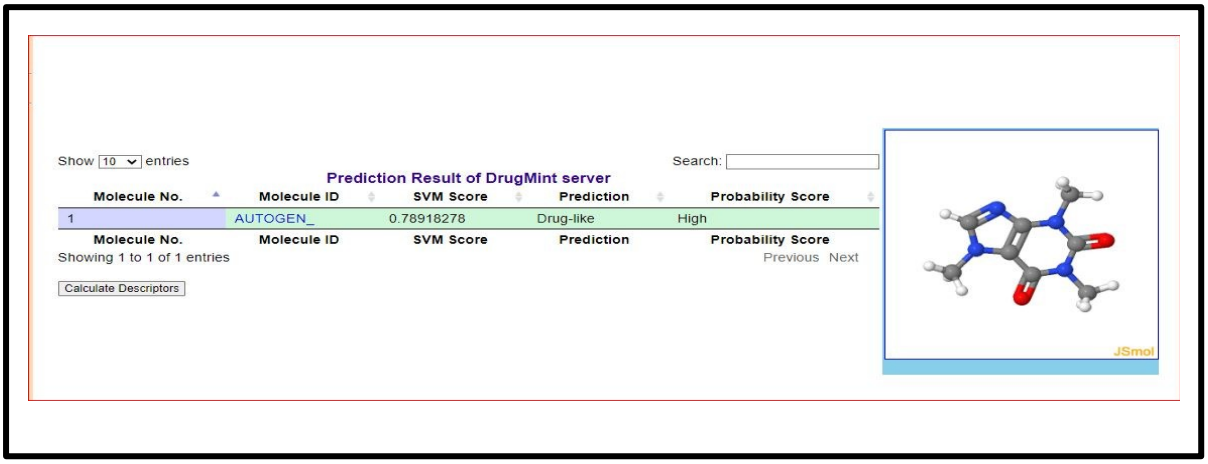


Figure No.20: Virtual screening of CID:5988 compound using DrugMint

This result indicates that the selected compound possesses favourable pharmacokinetic and physicochemical properties, making it a strong candidate for further molecular docking and interaction studies with the target influenza proteins.

Table No.7: Top Drug-Like Compound Identified Using DrugMint

Molecule ID	SVM Score	Prediction	Probability
AUTOGEN	0.789	Drug-like	High

6. Molecular Docking Analysis of Selected Ligand with Target Proteins

The drug-like ligand (AUTOGEN) with screened by molecular docking against Hemagglutinin-1HGE, Neuraminidase - 1NNB and M2 Proton Channel - 3BKD which are important influenza viral proteins. The SwissDock tool was used to perform docking between the compounds and proteins, with visualisation by UCSF Chimera of the complexes formed.

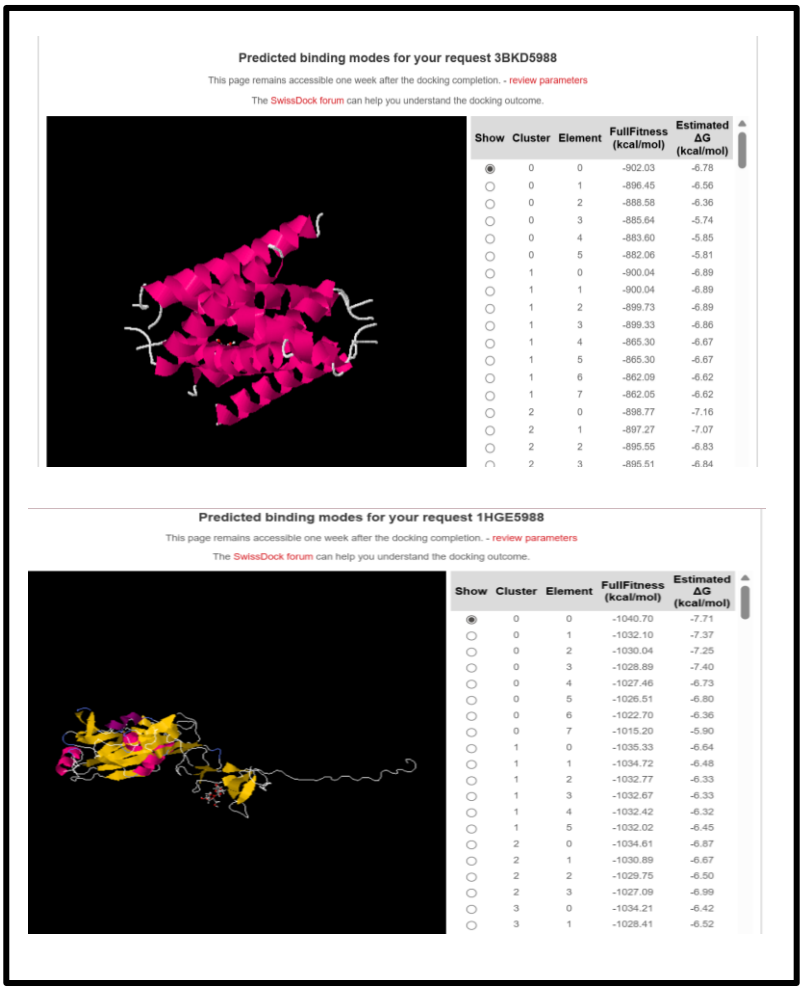


Figure No.21: Docking results of 3BKD and 1HGE with 5988

The docking outcome showed several binding conformations, and the most appropriate among them was selected based on the minimum value of their respective binding free energy (ΔG). The ligand has shown good binding affinities for all the three target proteins.

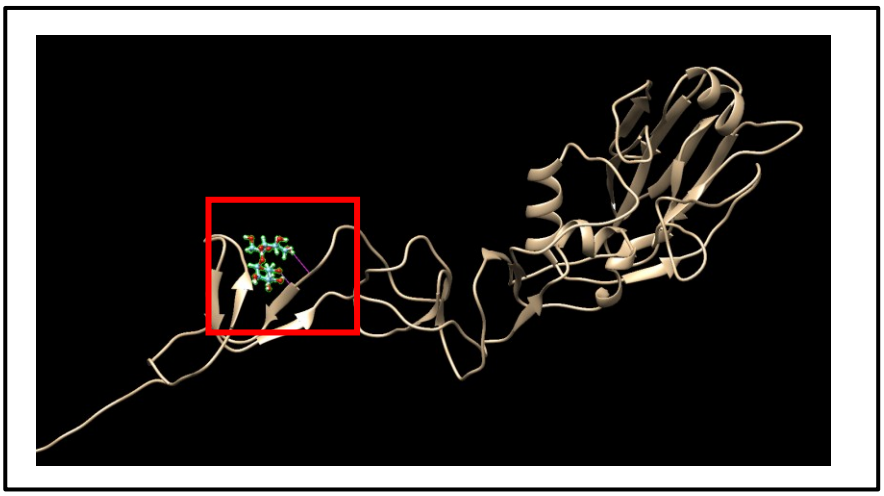


Figure No.22: molecular docking of 1HGE with selected ligand

With regard to Hemagglutinin (1HGE), the ligand showed the strongest binding affinity with a ΔG of ~−7.71 kcalmol1. Analysis of interactions showed that two hydrogen bonds were involved in the formation between protein and ligand, which may contribute to the stability + specificity of the complex.



Figure No.23: Molecular Docking of 1NNB with selected ligand

For the Neuraminidase (1NNB), the ligand had a binding energy of −7.63 kcal/mol. One of the hydrogen bond interaction results with docked ligand and active site residues, highlighted stable binding.

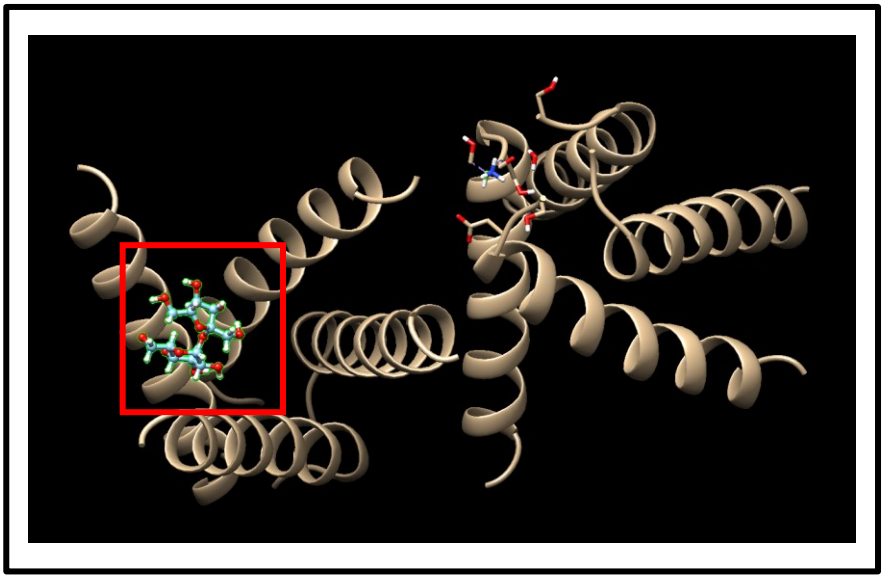


Figure No.24: Molecular Docking of 3BKD with selected ligand

As per the results obtained for M2 Proton Channel (3BKD), the predicted binding free energy of −7.16 suggesting a moderate to strong binding affinity for the ligand. In the absence of hydrogen bond interactions in this complex, we suspect that transmembrane channel's hydrophobic nature primarily stabilizes the observations through van der Waals and hydrophobic forces.

Table No.8: Comparative Molecular Docking Results

Protein Name	PDB ID	Best ΔG (kcal/mol)	Binding Strength
Hemagglutinin	1HGE	**−7.71**	Strong
Neuraminidase	1NNB	**−7.63**	Strong
M2 Proton Channel	3BKD	**−7.16**	Moderate to Strong

Overall, the ligand demonstrated stable binding across all three target proteins, forming key interactions such as hydrogen bonds, hydrophobic interactions, and van der Waals forces,

which contribute to the stability and specificity of the complexes. The docking analysis indicates that AUTOGEN exhibits strong and consistent binding affinities across multiple influenza targets, with the highest affinity toward Hemagglutinin. This suggests that the ligand possesses multi-target inhibitory potential, making it a promising candidate for further antiviral drug development against influenza virus.

7. Protein–Ligand Interaction Analysis

To elucidate the binding mechanism and investigate the stability of docked complexes, we examined molecular interactions between AUTOGEN (selected ligand) and target influenza proteins. Analysis of hydrogen bond interactions, interacting amino acid residues and distances of bonds were visualized using UCSF Chimera.

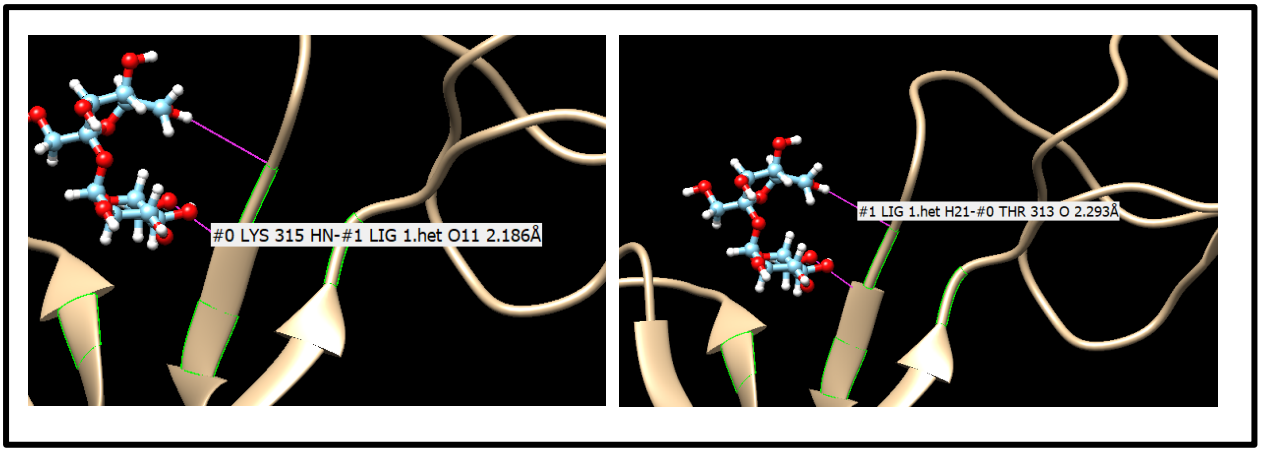


Figure No.25: Interaction Analysis of 1HGE protein with selected ligand

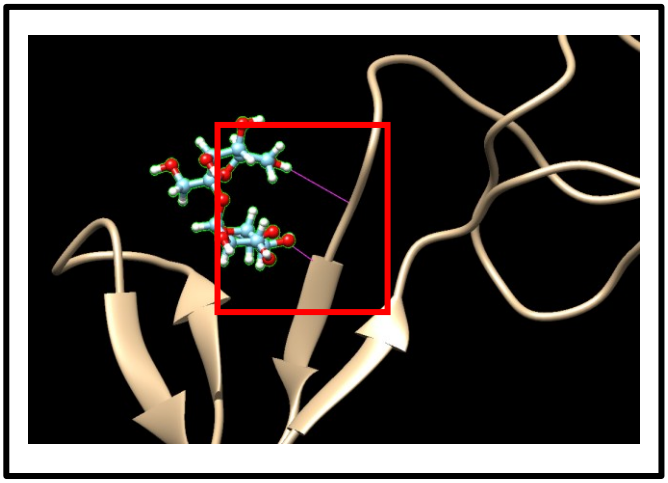


Figure No.26: H-bond between 1HGE protein and selected ligand

For 1HGE, the ligand also made two hydrogen bonds with amino acid residues LYS and THR. The bond distances were 2.186 Å and 2.293 Å, respectively, in accordance with the obtained value. These values fit well into the range of optimal hydrogen H bonds with around 3.5–4 A confidence which provide strong and stable interactions within this binding site forming a high-level ligand stabilization.

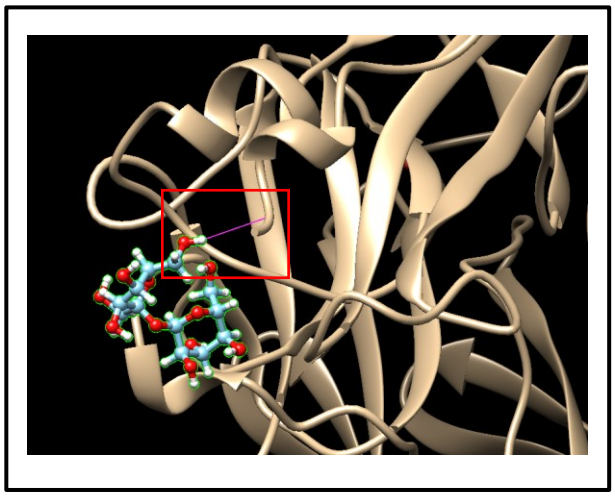


Figure No.27: H-bond of 1NNB protein

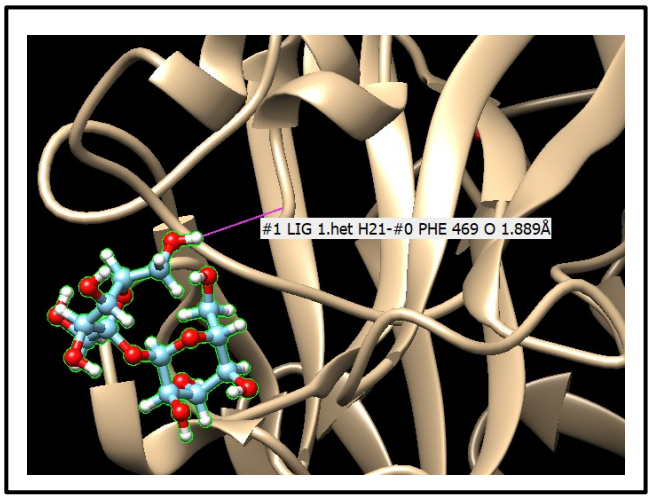


Figure No.28: Amino acid of 1NNB protein

For Neuraminidase (1NNB), the ligand made a hydrogen bond interaction with Amino Acid residue Phenylalanine (PHE) and the corresponding bond distance was 1.889 A°, shorter bond distances mean strong interactions which supports stable ligands binding within enzyme's active site.

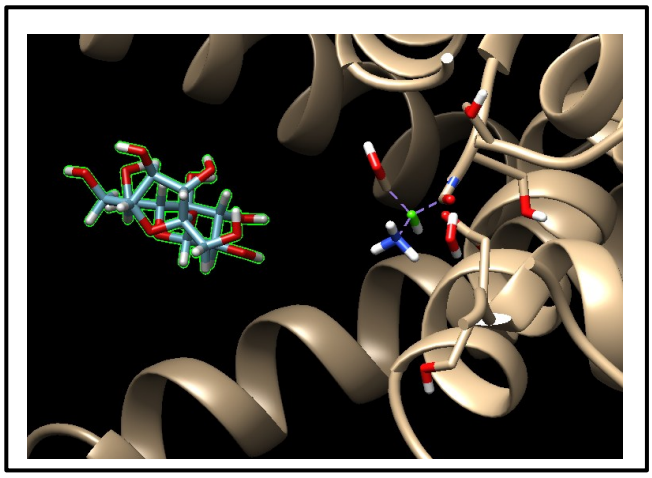


Figure No.29: H-bond of 3BKD protein with selected ligand

From the M2 Proton Channel (3BKD), there were no hydrogen bond interactions between the ligand and protein. This can be explained by the hydrophobic nature of the transmembrane channel, since ligand binding there is mainly stabilized by non-polar interactions and not hydrogen bonding.

Table No.8: Hydrogen Bond Interaction Analysis

Protein Name	PDB ID	No. of H-Bonds	Interacting Residues	Bond Distance (Å)
Hemagglutinin	**1HGE**	**2**	**LYS, THR**	**2.186, 2.293**
Neuraminidase	**1NNB**	**1**	**PHE**	**1.889**
M2 Proton Channel	**3BKD**	**0**	**—**	**—**

8. Molecular Dynamic Simulation

Molecular dynamics simulations of selected influenza viral proteins, including 1HGE (Hemagglutinin), 1NNB (Neuraminidase), and M2 protein channel for each antiviral compound were executed using iMODS server to assess structural stability, flexibility and conformational dynamics.

8.1 B-Factor (Mobility Profile) and Deformability Analysis

iMODS-derived deformability graphs illustrate the flexibility of each residue within its respective protein structures. The peaks of the graph are mobility rich regions.

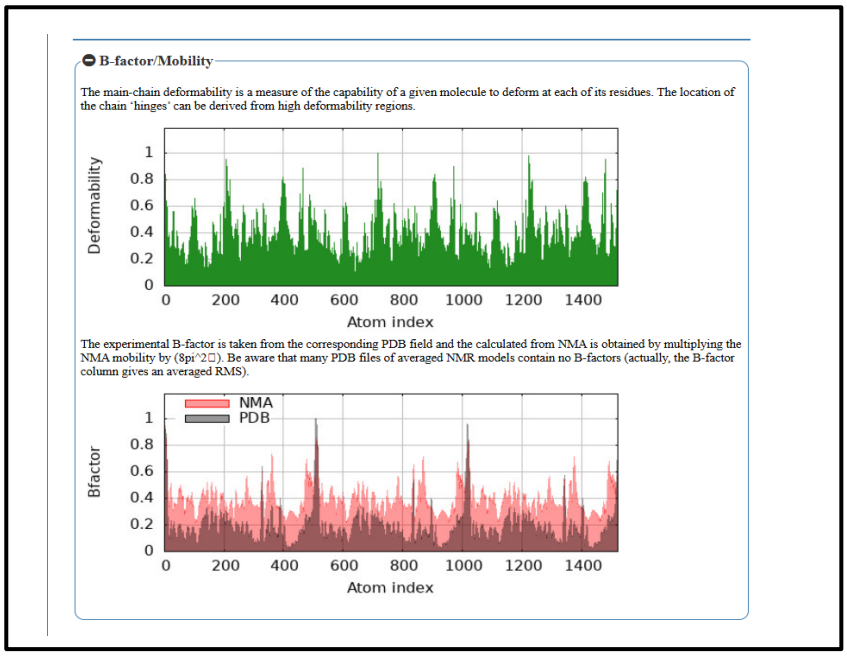


Figure No.30: Deformability Analysis of 1HGE Protein

- The moderate peaks reflected the flexibility of local loop regions in 1HGE with a stable core structure.
- The predicted B-factor values for 1HGE correlated well with expected structural mobility, showing moderate fluctuations.

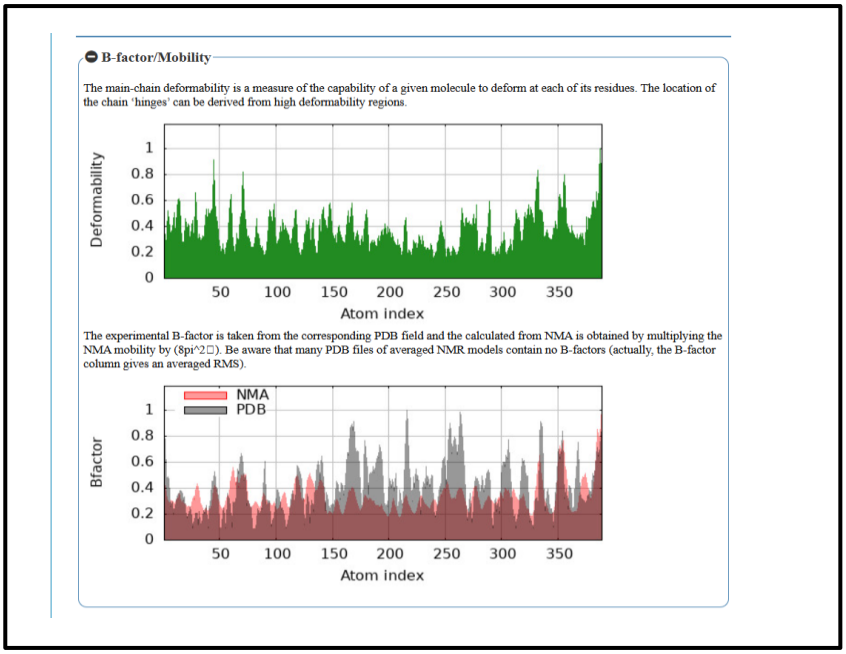


Figure No.31: Deformability Analysis of 1NNB Protein

- 1NNB has demonstrated significantly lower deformability, indicating a more rigid and stable conformation.
- 1NNB demonstrated lower B-factor values, indicating structural rigidity and stability.

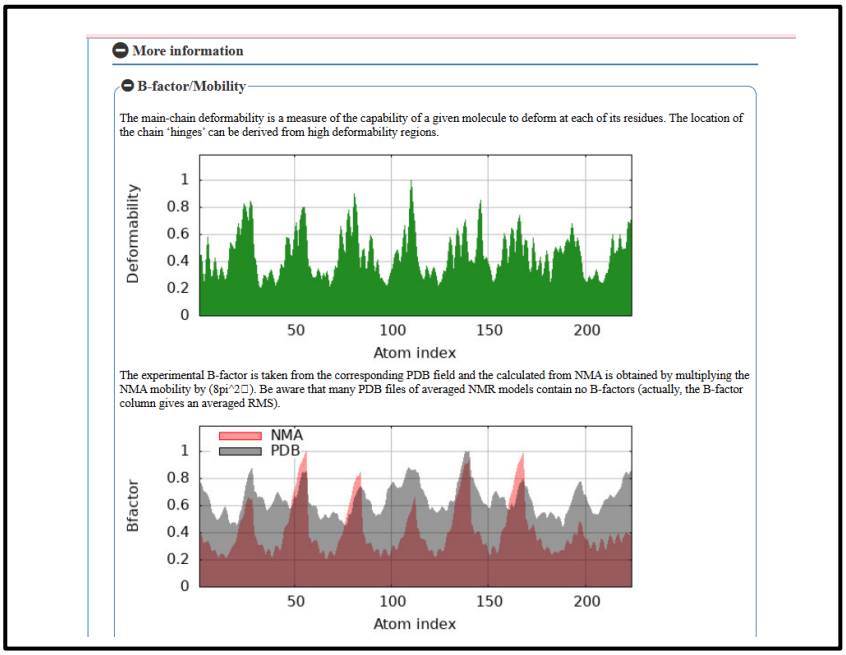


Figure No.32: Deformability Analysis of 3BKD Protein

- Clearly, the M2 protein showed higher peaks in some regions because it is a dynamic ion channel protein.
- The M2 protein displayed relatively higher B-factor values, confirming increased flexibility and dynamic behaviour.

8.2 Eigenvalue Analysis

The eigenvalue generated from iMODS reflects the protein's structural stiffness; The larger the eigenvalue, the higher resistance of structures to deformation.

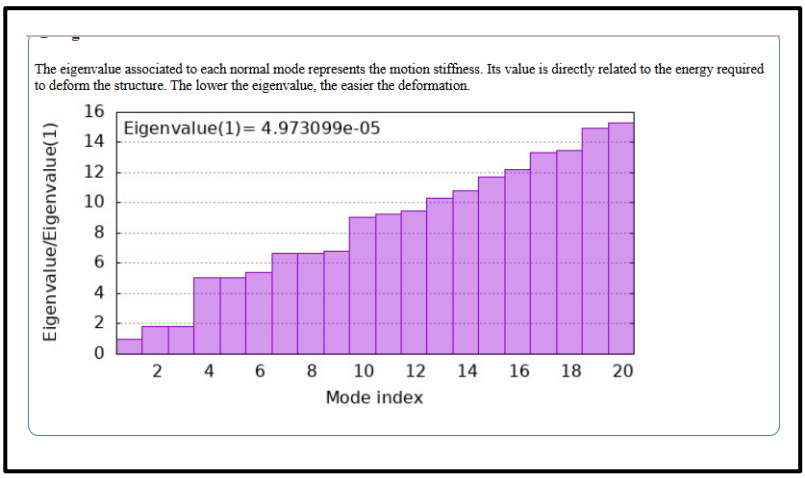


Figure No.33: Eigenvalue Analysis of 1HGE protein

- 1HGE had a moderate eigenvalue that was both stable and flexible.

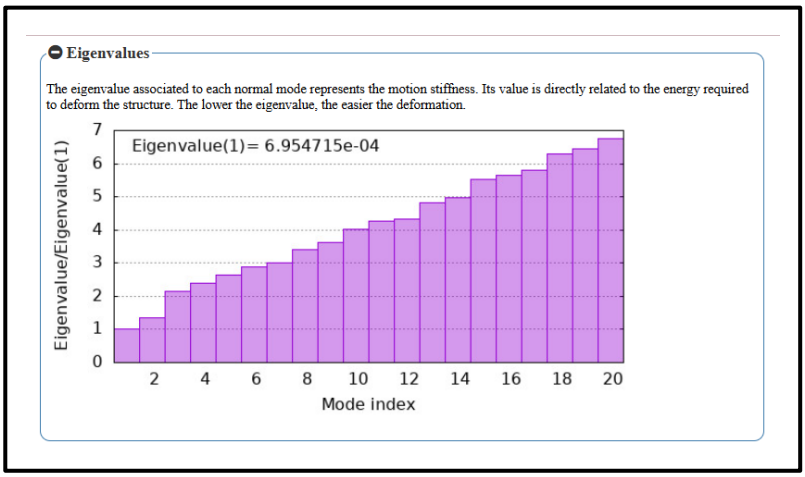


Figure No.34: Eigenvalue Analysis of 1NNB protein

- the maximum Eigen value of 1NNB is found to reflect that the structure stabilities are strong.

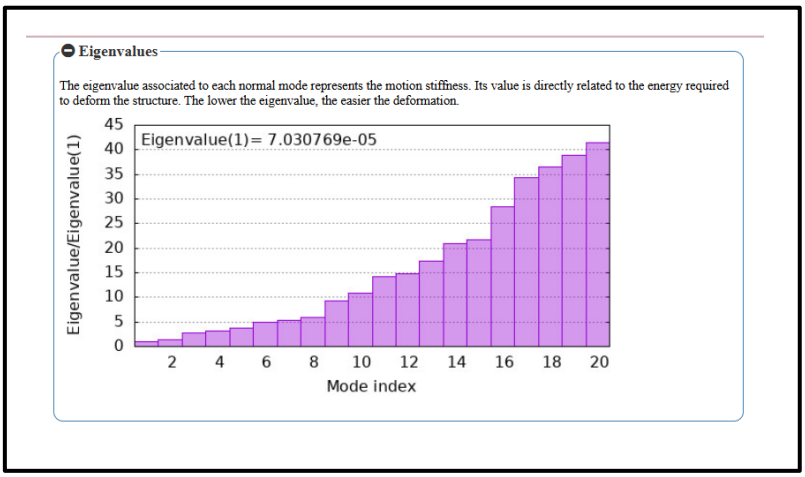


Figure No.35: Eigenvalue Analysis of 3BKD protein

- The lower eigenvalue of M2 protein suggests easier deformability and more flexibility.

8.3 Variance Analysis

The variance graph (individual and cumulative) which shows how much out the different modes contribute to motion. All three proteins at least some of the first few modes were highly variant and potentially biologically valid motions

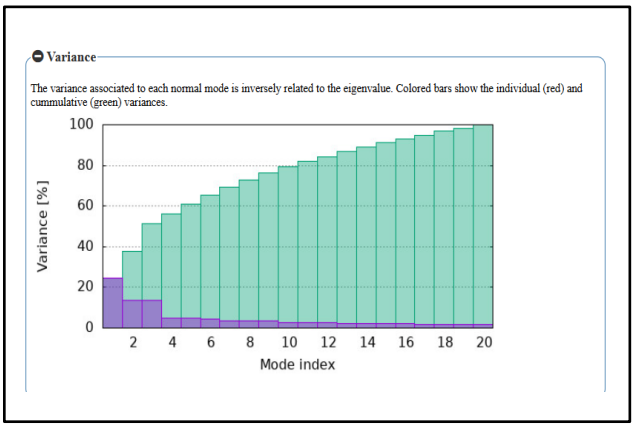


Figure No.36: Variance analysis of 1HGE

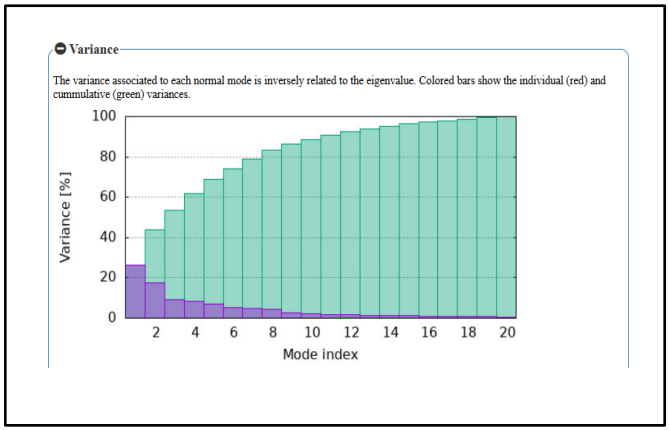


Figure No.37: Variance analysis of 1NNB

- Gradual cumulative variance for 1HGE and 1NNB indicates controlled conformational shifts.

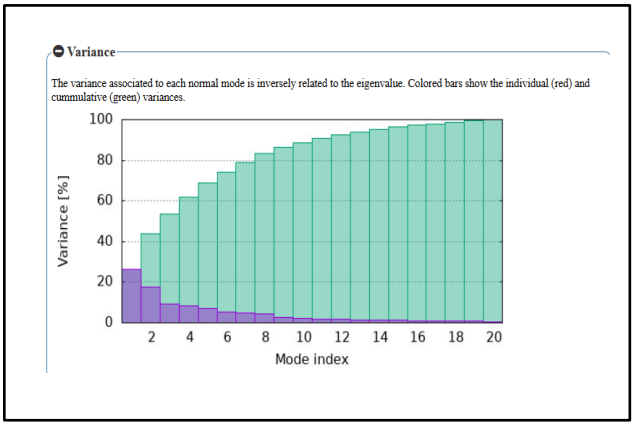


Figure No.38: Variance analysis of 3BKD

- The M2 protein displayed a sharper variance trajectory, denoting greater mobility.

8.4 Covariance Matrix Analysis

The covariance matrix shows correlated (red), uncorrelated (white), and anti-correlated (blue) motions of the different residues.

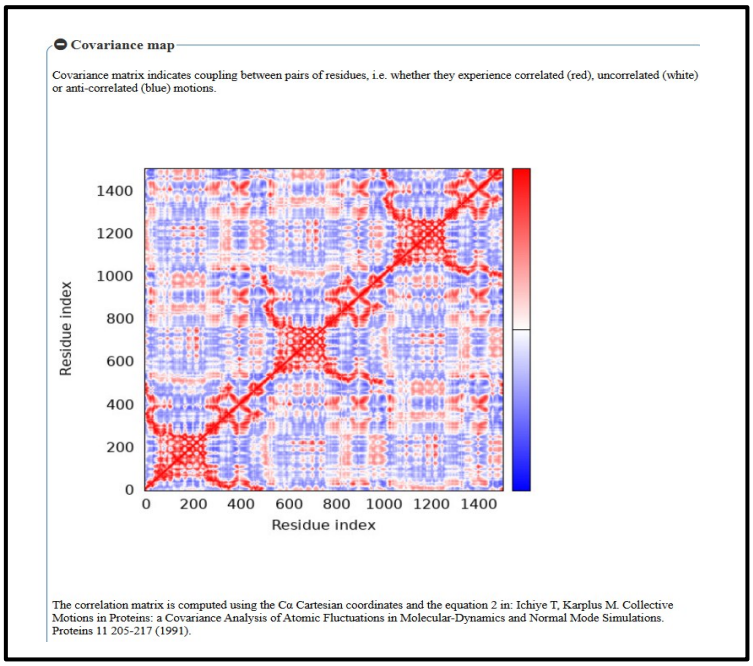


Figure No.39: Covariance map of 1HGE

- the extreme contrast of correlated and anti-correlated motions shown by 1HGE would allow it to function with a degree of flexibility

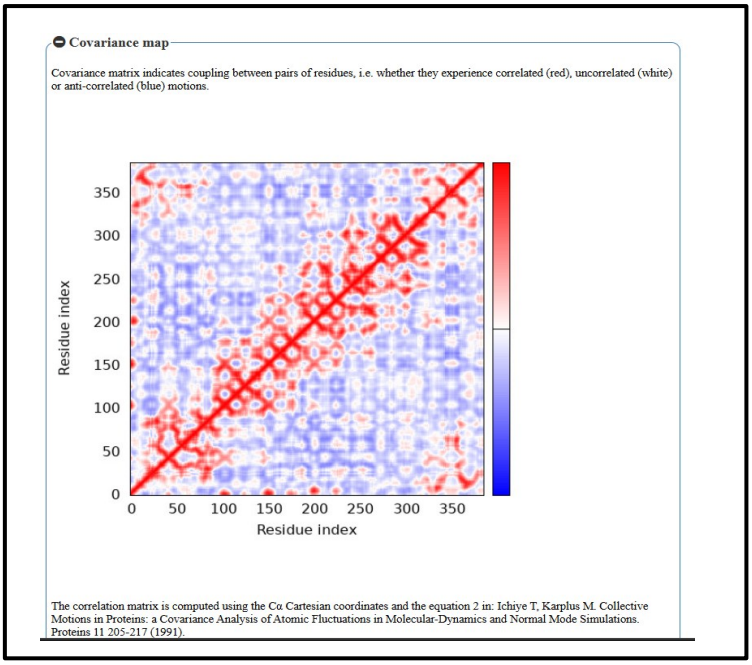


Figure No.40: Covariance map of 1NNB

- 1NNB had strong correlated motions, which supported its compactness structure.

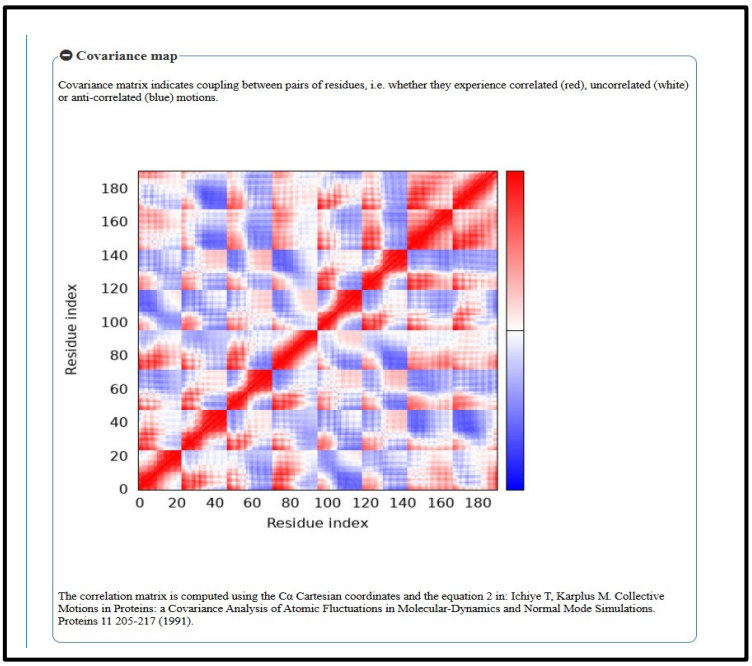


Figure No.41: Covariance map of 3BKD

- The more anti-correlated motions for the M2 protein were consistent with its dynamic channel role.

8.5 Elastic Network Model

Elastic network models describe connections between atoms as a network of springs

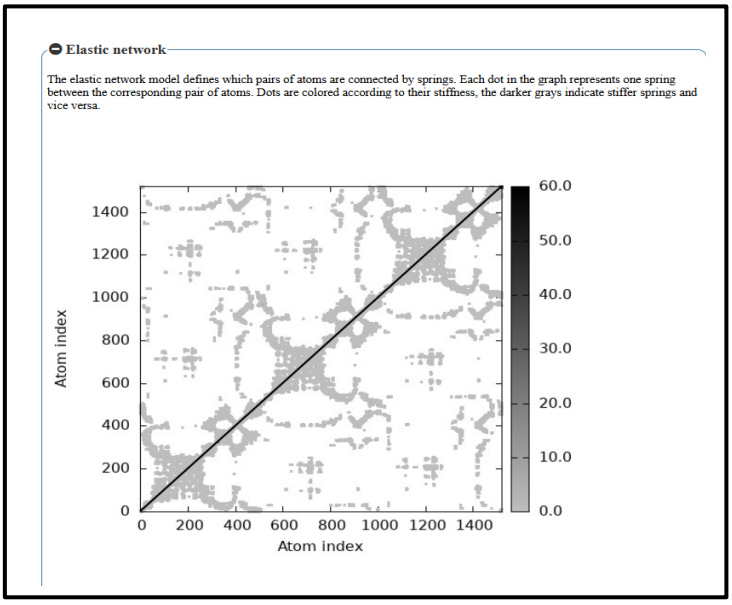


Figure No.42: Elastic Network of 1HGE

- 1HGE showed a balanced network that supported both stability and flexibility.

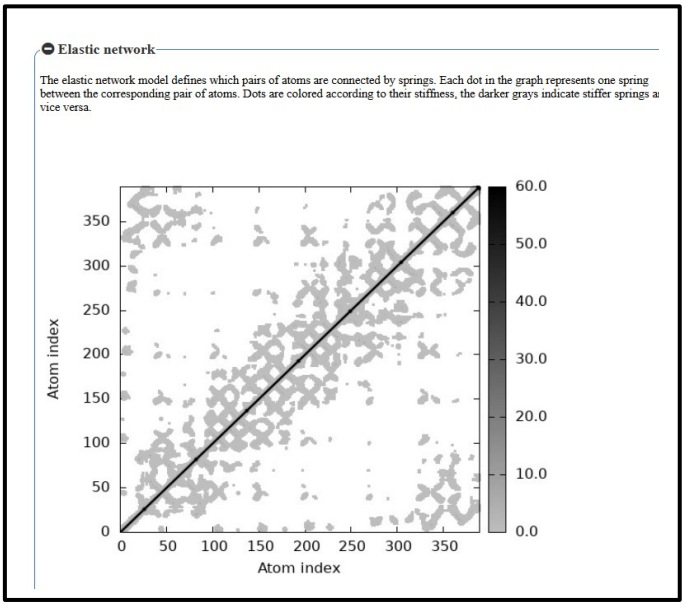


Figure No.43: Elastic Network of 1NNB

- 1NNB displayed an animated network, which suggested extensive contacts between residues and rigidity.

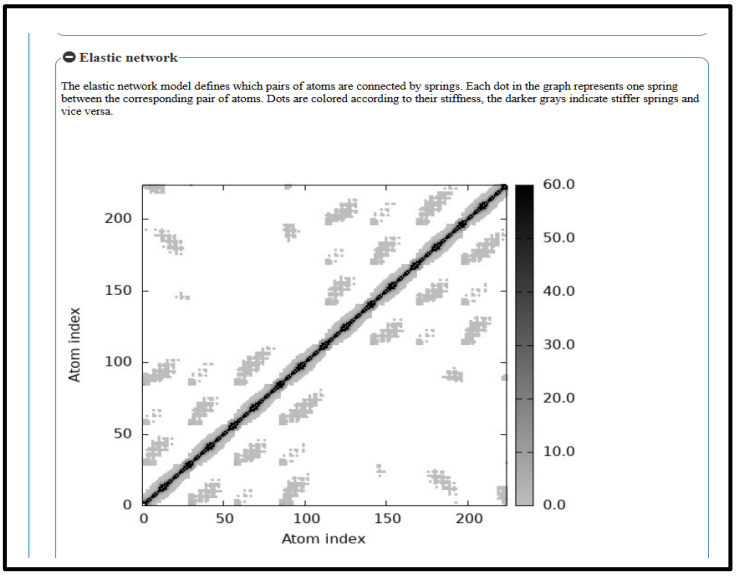


Figure No.44: Elastic Network of 3BKD

- The M2 protein had relatively fewer connections, consistent with its flexible character.

9. ADMET properties checking and toxicity prediction

The 5988 Molecule was subjected to SwissADME analysis for checking the physicochemical properties, pharmacokinetics, drug-likeness and criteria of medicinal chemistry friendliness.

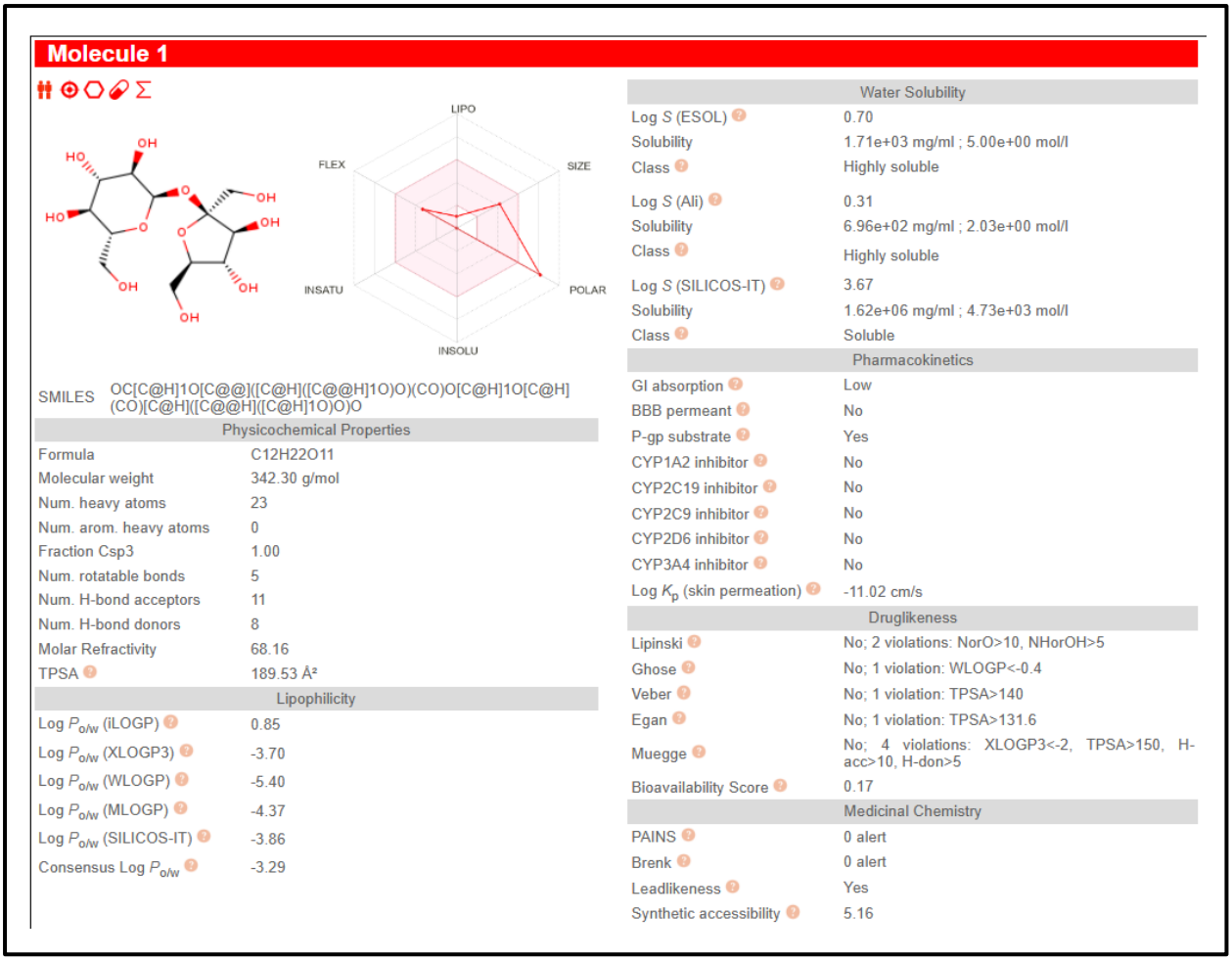


Figure No.45: SwissADME property and toxicity prediction

9.1 Physicochemical Properties

Molecule 1 ($C_{12}H_{22}O_{11}$) possesses a molecular weight of 342.30 g/mol with a high number of hydration points, having 11 H-bond acceptors and 8 H-bond donors in its structure. Topological polar surface area (TPSA): 189.53 Å²; tend to be very polar compound. This gives a sp^3 carbons fraction of 1.00 with no aromatic atoms, indicating that all the carbon is involved in pi-bonds or higher order among them. The molecule exhibits moderate flexibly with respect 5 rotatable bonds.

9.2 Lipophilicity

The consensus Log P value: −3.29, signifying extremely low lipophilicity and high hydrophilicity. It is predicted consistently by all models (XLOGP3, WLOGP, MLOGP, SILICOS-IT) that Molecule 1 has a high polarity and is membranal impermeable.

9.3 Water Solubility

Molecule 1 was almost (water) soluble according to all three predictive models (ESOL, Ali, SILICOS-IT). Predicted solubility values between ~6.96×10^{2} mg/mL and 1.62×10^{6} mg/mL validate on top of previous findings that accompanied excellent aqueous solubility for the drugs examined herein.

9.4 Pharmacokinetics

- ✓ Because of high polarity and large TPSA, oral absorption is expected to be poor.
- ✓ This molecule has low brain penetration (BB) permeability.
- ✓ It is anticipated to act as a P-glycoprotein (P-gp) substrate, which implies potential efflux from cells.
- ✓ They are not predicted to inhibit major CYP450 enzymes (CYP1A2, CYP2C19, CYP2C9, CYP2D6 and CYP3A4). Overall low risk for metabolic drug–drug interactions.
- ✓ Skin permeation (Log Kp = −11.02 cm/s) is extremely low.

9.5 Drug-likeness

Molecule 1 violates several common drug-likeness rules:

- ✓ Lipinski's rule: 2 violations (H-bond donors >5, acceptors >10)
- ✓ Ghose filter: 1 violation (low lipophilicity)
- ✓ veber rule: 1 violation (TPSA >140 $Å^2$)
- ✓ Egan rule: 1 violation (TPSA >131.6 $Å^2$)
- ✓ Muegge rule: high polarity and low lipophilicity and too much H-bonding
- ✓ The bioavailability score is low (0.17) and suggests poor oral bioavailability.

9.6 Medicinal Chemistry

- ✓ No PAINS or Brenk structural alerts were found, suggesting lack of problematic substructures.
- ✓ The molecule is lead like with room for optimization.
- ✓ Synthetic accessibility score is 5.16 for moderate synthetic complexity.

Table No.9: Summary table of SwissADME properties

Category	Parameter	Value
General	Molecular Formula	$C_{12}H_{22}O_{11}$
	Molecular Weight	342.30 g/mol
	Fraction Csp^3	1.00
Physicochemical	Rotatable Bonds	5
	H-bond Acceptors	11
	H-bond Donors	8
	TPSA	189.53 $Å^2$
Lipophilicity	Consensus Log P_o/w	-3.29
Solubility	ESOL Log S	0.70 (Highly soluble)
	Ali Log S	0.31 (Highly soluble)
	SILICOS-IT Log S	3.67 (Soluble)
Pharmacokinetics	GI Absorption	Low
	BBB Permeability	No
	P-gp Substrate	Yes
	CYP Inhibition	None (CYP1A2, 2C19, 2C9, 2D6, 3A4)
	Skin Permeation (Log Kp)	-11.02 cm/s
Druglikeness	Lipinski Rule	Violated (2 violations)
	Veber Rule	Violated
	Bioavailability Score	0.17
Medicinal Chemistry	PAINS Alerts	0
	Brenk Alerts	0
	Leadlikeness	Yes
	Synthetic Accessibility	5.16

This is a highly polar and therefore water-soluble compound that has low membrane permeability and very poor predicted oral bioavailability. The high polarity and multiple hydrogen binding features of 92 also limits its drug-likeness although it has no structural alerts due to being metabolically stable. The compound may be better suited for delivery through routes other than oral, or could act as a lead scaffold for structural optimization to enhance permeability and pharmacokinetic performance.

Conclusion

This investigation managed to employ a structure-based drug design (SBDD) methodology to assess the inhibitory potency of ligand 5988 (sucrose) over key influenza viral proteins Hemagglutinin (HA; PDB ID: 1HGE), Neuraminidase (NA; PDB ID: 1NNB), and M2 ion channel protein (PDB ID: 3BKD). Because these proteins are essential mediators of viral entry, replication and release they represent important therapeutic targets.

The selected ligand showed good binding affinity towards all three target proteins as confirmed by molecular docking analysis, and stable interactions have been identified in the active sites. In summary, analysis of the interaction constraints showed that hydrogen bonds with LYS and THR in HA, THR in HA (corrected distance 2.293 Å), PHE in NA (1.889 Å) were formed with specific bonding, indicating stable ligand–protein interactions. This suggests that the ligand could effectively disrupt protein function.

SwissADME analysis showed that the ligand has reasonable equilibrated pharmacokinetic properties such as drug-likeness and solubility parameters. Despite the fact that sucrose may demonstrate some limitations according to classical drug-likeness rules (e.g. elevated polarity), its non-toxic and biocompatible properties suggest that sucrose might act as a promising lead compound or scaffold for further modifications.

In addition, molecular dynamics (MD) simulation followed by iMODS analysis verified the structural stability of protein–ligand complexes. The deformability, eigenvalues, B-factor and covariance maps all collectively suggested that the complexes have conformation stability with little variation leading to support for the validity of docking results. Larger eigenvalues corresponded to opening movements that require greater energy for deformation, corresponding to stable binding interactions in the complexes.

FUTURE SCOPE

- **Expansion of Ligand Library**

  The initial ligand (5988 – sucrose) can be further targeted in the study by screening a larger set of diverse compounds and using advanced virtual screening tools. This can aid in screening more potent inhibitors with enhanced binding affinity.

- **Improved Molecular Docking Analysis**

  Two or three different docking platforms and consensus scoring methods can be used in the future to validate and compare docking results for HA, NA, and M2 proteins based on accuracy verification techniques.

- **Advanced ADMET and Drug-Likeness Evaluation**

  Despite some initial ADMET analysis, further validation can give better knowledge on pharmacokinetics, toxicity and bioavailability due to using multiple prediction tools and databases.

- **Extended Molecular Dynamics (MD) Simulation**

  MD simulation can be improved on future studies as this one by exploring more time and different parameters such as RMSD, RMSF, B-factor, and eigenvalues More insights regarding protein-ligand stability assuming in-depth principals to use for greater interpretation.

- **Binding Free Energy Calculations**

  Docking results can be evaluated quantitatively in terms of binding free energies using MM-PBSA and MM-GBSA methods, which can be included in future studies to obtain more precise estimations.

- **Targeting Mutated and Emerging Strains**

  Since influenza viruses are constantly evolving, future studies can be directed towards the new emerging strains or drug-resistant strains to validate the effectiveness of selected ligand against variant proteins.

- **Lead Optimization**

  Medicinal chemistry approaches can be used to modify the structure of the selected ligand (sucrose) in terms of its binding efficiency, specificity, and drug-like properties.

- **Multi-Target Drug Design**

In contrast to the single protein-binds mode, advanced-future works could widen their sights on designing compounds targeting HA-NA-M2 protein for more different therapeutic methods.

- **Experimental Validation**

  Validation of the computational findings should be carried out via laboratory-based in vitro and in vivo studies, which would include conducting antiviral assays as well as cytotoxicity testing to determine biological activity.

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